TalentLens: Recruitment & Onboarding Codebook

This analyses of recruitment and onboarding data is from Pegasus Mortgage Lending Center, The data set is used to explore how Business Analytics can improve the efficiency and effectiveness of the organisation's recruitment process.

The dataset contains 31,500 unique candidate records from the 2020 - 2026 covering information such as candidate characteristics, recruitment sources, progression through recruitment stages, communication activities, licensing information and hiring outcomes.

By applying descriptive analysis, exploratory data analysis, recruitment funnel analysis and predictive modelling, the dissertation aims to identify patterns in candidate progression, understand where candidates are lost during the recruitment process, and assess whether historical data can support better decision-making.

The analysis also considers important data-quality issues, including substantial missingness and inconsistencies between recruitment-status fields, to ensure that the findings are interpreted responsibly. Ultimately, the purpose of the dissertation is to provide evidence-based and practical recommendations that can help Pegasus Mortgage Lending Center improve recruitment processes, while recognising the limitations and risks associated with using historical recruitment data.

In [1]:
```
# Import the libraries and the dataset

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.metrics import silhouette_score
from sklearn.cluster import KMeans
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_val_score

import scipy.cluster.hierarchy as sch
from sklearn.cluster import AgglomerativeClustering, KMeans

from sklearn.decomposition import PCA
```

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, precision_score, recall_score, confusion_matrix, classification_report

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    accuracy_score, classification_report, confusion_matrix,
    average_precision_score, f1_score
)

RANDOM_STATE = 42
```

In [3]:
```
import os
print(os.getcwd())
```

```
C:\Users\REGAL COMPUTER\OneDrive\Desktop\Dissertation\Final Submission
```

In [2]:
```
import warnings
warnings.filterwarnings('ignore')
```

In [4]:
```
#Load the dataset
df = pd.read_csv("C:\\Users\\REGAL COMPUTER\\OneDrive\\Desktop\\Dissertation\\Full Draft\\Clean Dataset.csv")
```

Data Preparation

In [5]:
```
df.head()
```

Out[5]:

	Candidate ID.1	Gender	Email	Phone Number	Mobile Number	City	Country	Created Time	Modified Time	Currency	...	Work Type	Licensi Stat
0	ZR_28480_CAND	Male	Yes	No	Yes	NORTH YORK	Canada	12/29/2023 9:31	3/6/2026 13:01	CAD	...	NaN	Authoriz to S
1	ZR_28476_CAND	Female	Yes	Yes	No	Cambridge	Canada	12/20/2023 16:31	5/6/2026 14:43	CAD	...	Full Time	Expir
2	ZR_28475_CAND	Male	Yes	No	Yes	Brampton	Canada	12/20/2023 16:29	5/6/2026 14:43	CAD	...	NaN	N
3	ZR_28474_CAND	Male	Yes	No	Yes	Toronto	Canada	12/19/2023 8:54	5/6/2026 14:43	CAD	...	Part Time	N
4	ZR_28473_CAND	Male	Yes	No	Yes	Cambridge	Canada	12/19/2023 8:54	5/6/2026 14:43	CAD	...	NaN	Authoriz to S

5 rows × 45 columns

In [6]:
```
df.shape
```

Out[6]:
```
(1048575, 45)
```

In [7]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1048575 entries, 0 to 1048574
Data columns (total 45 columns):
 #   Column                                  Non-Null Count   Dtype
---  ------                                  --------------   -----
 0   Candidate ID.1                          31500 non-null   str
 1   Gender                                  31500 non-null   str
 2   Email                                   31500 non-null   str
 3   Phone Number                            31500 non-null   str
 4   Mobile Number                           31500 non-null   str
 5   City                                    9361 non-null    str
 6   Country                                 31500 non-null   str
 7   Created Time                            31500 non-null   str
 8   Modified Time                           31500 non-null   str
 9   Currency                                31500 non-null   str
 10  Last Activity Time                      31500 non-null   str
 11  Last emailed                            11228 non-null   str
 12  Source                                  31500 non-null   str
 13  Email Opt Out                           31500 non-null   str
 14  Is Hot Candidate                        31500 non-null   str
 15  Is Locked                               31500 non-null   str
 16  Candidate Status                        31500 non-null   str
 17  Initial Contact Date                    31500 non-null   str
 18  Institute 1 Sign Up Date                5458 non-null    str
 19  Hire Date                               1716 non-null    str
 20  Ethnicity                               31500 non-null   str
 21  Religion                                31500 non-null   str
 22  Position Category                       31500 non-null   str
 23  License Class                           17628 non-null   str
 24  License year                            22971 non-null   float64
 25  Years of Experience                     22972 non-null   float64
 26  Current and Previous Mortgage           13694 non-null   str
 27  Equifax Credit Score                    31498 non-null   str
 28  Criminal History                        1401 non-null    str
 29  Prior Discipline Action by a Regulator  8 non-null       str
 30  Professional License Suspension         1508 non-null    str
 31  Candidate Stage                         27914 non-null   str
 32  Realtor License                         1452 non-null    str
 33  Hiring Validation                       31500 non-null   str
 34  1 Month Fee Waiver                      1420 non-null    str
 35  Work Type                               1191 non-null    str
 36  Licensing Status                        19285 non-null   str
```

```
 37  Province                          31137 non-null   str
 38  Parked                            31419 non-null   str
 39  Pre-Interview Info Email          31419 non-null   str
 40  Post Interview Info Email         31419 non-null   str
 41  Discussion Type                   1990 non-null    str
 42  Live Transfer                     1989 non-null    str
 43  Discussion Stage                  902265 non-null  str
 44  Schedule Discussion               31419 non-null   str
dtypes: float64(2), str(43)
memory usage: 377.7 MB
```

In [8]:
```python
#Converting Dates from string to Datetime object
date_columns = [
'Created Time',
'Modified Time',
'Hire Date',
'Initial Contact Date'
]

for col in date_columns:
    df[col] = pd.to_datetime(df[col], errors='coerce')
```

In [9]:
```python
# Encoding Yes & No Columns to Binary code
yes_no_columns = ['Email',
                  'Phone Number',
                  'Mobile Number',
                  'Email Opt Out',
                  'Is Locked',
                  'Equifax Credit Score',
                  'Hiring Validation',
                  'Schedule Discussion',
                  'Parked',
                  'Pre-Interview Info Email',
                  'Post Interview Info Email'
                 ]

for col in yes_no_columns:
    df[col + '_Encoded'] = df[col].map({
        'Yes': 1,
        'No': 0
    })
```

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1048575 entries, 0 to 1048574
Data columns (total 56 columns):
 #   Column                                  Non-Null Count   Dtype
---  ------                                  --------------   -----
 0   Candidate ID.1                          31500 non-null   str
 1   Gender                                  31500 non-null   str
 2   Email                                   31500 non-null   str
 3   Phone Number                            31500 non-null   str
 4   Mobile Number                           31500 non-null   str
 5   City                                    9361 non-null    str
 6   Country                                 31500 non-null   str
 7   Created Time                            31500 non-null   datetime64[us]
 8   Modified Time                           31500 non-null   datetime64[us]
 9   Currency                                31500 non-null   str
 10  Last Activity Time                      31500 non-null   str
 11  Last emailed                            11228 non-null   str
 12  Source                                  31500 non-null   str
 13  Email Opt Out                           31500 non-null   str
 14  Is Hot Candidate                        31500 non-null   str
 15  Is Locked                               31500 non-null   str
 16  Candidate Status                        31500 non-null   str
 17  Initial Contact Date                    23959 non-null   datetime64[us]
 18  Institute 1 Sign Up Date                5458 non-null    str
 19  Hire Date                               1716 non-null    datetime64[us]
 20  Ethnicity                               31500 non-null   str
 21  Religion                                31500 non-null   str
 22  Position Category                       31500 non-null   str
 23  License Class                           17628 non-null   str
 24  License year                            22971 non-null   float64
 25  Years of Experience                     22972 non-null   float64
 26  Current and Previous Mortgage           13694 non-null   str
 27  Equifax Credit Score                    31498 non-null   str
 28  Criminal History                        1401 non-null    str
 29  Prior Discipline Action by a Regulator  8 non-null       str
 30  Professional License Suspension         1508 non-null    str
 31  Candidate Stage                         27914 non-null   str
 32  Realtor License                         1452 non-null    str
 33  Hiring Validation                       31500 non-null   str
 34  1 Month Fee Waiver                      1420 non-null    str
 35  Work Type                               1191 non-null    str
 36  Licensing Status                        19285 non-null   str
```

```
 37  Province                                31137 non-null   str
 38  Parked                                  31419 non-null   str
 39  Pre-Interview Info Email                31419 non-null   str
 40  Post Interview Info Email               31419 non-null   str
 41  Discussion Type                         1990 non-null    str
 42  Live Transfer                           1989 non-null    str
 43  Discussion Stage                        902265 non-null  str
 44  Schedule Discussion                     31419 non-null   str
 45  Email_Encoded                           31500 non-null   float64
 46  Phone Number_Encoded                    31500 non-null   float64
 47  Mobile Number_Encoded                   31500 non-null   float64
 48  Email Opt Out_Encoded                   31500 non-null   float64
 49  Is Locked_Encoded                       31500 non-null   float64
 50  Equifax Credit Score_Encoded            31498 non-null   float64
 51  Hiring Validation_Encoded               31500 non-null   float64
 52  Schedule Discussion_Encoded             31419 non-null   float64
 53  Parked_Encoded                          31419 non-null   float64
 54  Pre-Interview Info Email_Encoded        31419 non-null   float64
 55  Post Interview Info Email_Encoded       31419 non-null   float64
dtypes: datetime64[us](4), float64(13), str(39)
memory usage: 464.0 MB
```

In [11]:
```
#Checking Missing Values
df.isnull().sum()
```

Out[11]:

Candidate ID.1	1017075
Gender	1017075
Email	1017075
Phone Number	1017075
Mobile Number	1017075
City	1039214
Country	1017075
Created Time	1017075
Modified Time	1017075
Currency	1017075
Last Activity Time	1017075
Last emailed	1037347
Source	1017075
Email Opt Out	1017075
Is Hot Candidate	1017075
Is Locked	1017075
Candidate Status	1017075
Initial Contact Date	1024616
Institute 1 Sign Up Date	1043117
Hire Date	1046859
Ethnicity	1017075
Religion	1017075
Position Category	1017075
License Class	1030947
License year	1025604
Years of Experience	1025603
Current and Previous Mortgage	1034881
Equifax Credit Score	1017077
Criminal History	1047174
Prior Discipline Action by a Regulator	1048567
Professional License Suspension	1047067
Candidate Stage	1020661
Realtor License	1047123
Hiring Validation	1017075
1 Month Fee Waiver	1047155
Work Type	1047384
Licensing Status	1029290
Province	1017438
Parked	1017156
Pre-Interview Info Email	1017156
Post Interview Info Email	1017156
Discussion Type	1046585

```
Live Transfer                          1046586
Discussion Stage                        146310
Schedule Discussion                    1017156
Email_Encoded                          1017075
Phone Number_Encoded                   1017075
Mobile Number_Encoded                  1017075
Email Opt Out_Encoded                  1017075
Is Locked_Encoded                      1017075
Equifax Credit Score_Encoded           1017077
Hiring Validation_Encoded              1017075
Schedule Discussion_Encoded            1017156
Parked_Encoded                         1017156
Pre-Interview Info Email_Encoded       1017156
Post Interview Info Email_Encoded      1017156
dtype: int64
```

In [12]:
```
df = df[df["Candidate ID.1"].notna()].copy()  # keep rows with candidate ID only
```

In [13]:
```
df.isnull().sum()
```

Out[13]:

Candidate ID.1	0
Gender	0
Email	0
Phone Number	0
Mobile Number	0
City	22139
Country	0
Created Time	0
Modified Time	0
Currency	0
Last Activity Time	0
Last emailed	20272
Source	0
Email Opt Out	0
Is Hot Candidate	0
Is Locked	0
Candidate Status	0
Initial Contact Date	7541
Institute 1 Sign Up Date	26042
Hire Date	29784
Ethnicity	0
Religion	0
Position Category	0
License Class	13872
License year	8529
Years of Experience	8528
Current and Previous Mortgage	17806
Equifax Credit Score	2
Criminal History	30099
Prior Discipline Action by a Regulator	31492
Professional License Suspension	29992
Candidate Stage	3586
Realtor License	30048
Hiring Validation	0
1 Month Fee Waiver	30080
Work Type	30309
Licensing Status	12215
Province	363
Parked	81
Pre-Interview Info Email	81
Post Interview Info Email	81
Discussion Type	29510

```
Live Transfer                          29511
Discussion Stage                       29512
Schedule Discussion                       81
Email_Encoded                              0
Phone Number_Encoded                       0
Mobile Number_Encoded                      0
Email Opt Out_Encoded                      0
Is Locked_Encoded                          0
Equifax Credit Score_Encoded               2
Hiring Validation_Encoded                  0
Schedule Discussion_Encoded               81
Parked_Encoded                            81
Pre-Interview Info Email_Encoded          81
Post Interview Info Email_Encoded         81
dtype: int64
```

In [14]:
```
categorical_columns = df.select_dtypes(include='object').columns

for col in categorical_columns:
    df[col] = df[col].fillna('Unknown')
```

In [15]:
```
numerical_columns = df.select_dtypes(include=['int64', 'float64']).columns

for col in numerical_columns:
    df[col] = df[col].fillna(df[col].median())
```

In [16]:
```
date_columns = ['Created Time', 'Modified Time']

for col in date_columns:
    df[col] = df[col].fillna(df[col].mode()[0])
```

In [17]:
```
df.replace(r'^\s*$', np.nan, regex=True, inplace=True)
```

Out[17]:

	Candidate ID.1	Gender	Email	Phone Number	Mobile Number	City	Country	Created Time	Modified Time	Currency	...	I Number_En
0	ZR_28480_CAND	Male	Yes	No	Yes	NORTH YORK	Canada	2023-12-29 09:31:00	2026-03-06 13:01:00	CAD	...	
1	ZR_28476_CAND	Female	Yes	Yes	No	Cambridge	Canada	2023-12-20 16:31:00	2026-05-06 14:43:00	CAD	...	
2	ZR_28475_CAND	Male	Yes	No	Yes	Brampton	Canada	2023-12-20 16:29:00	2026-05-06 14:43:00	CAD	...	
3	ZR_28474_CAND	Male	Yes	No	Yes	Toronto	Canada	2023-12-19 08:54:00	2026-05-06 14:43:00	CAD	...	
4	ZR_28473_CAND	Male	Yes	No	Yes	Cambridge	Canada	2023-12-19 08:54:00	2026-05-06 14:43:00	CAD	...	
...	...	...	...	...	...	...	...	...	...	...	...	
31495	ZR_28486_CAND	Male	Yes	No	Yes	Brampton	Canada	2024-01-02 12:13:00	2025-03-10 16:30:00	CAD	...	
31496	ZR_28485_CAND	Female	Yes	Yes	No	Brampton ,ON	Canada	2024-01-02 12:13:00	2026-02-19 11:11:00	CAD	...	
31497	ZR_28484_CAND	Male	Yes	No	Yes	Richmond Hill	Canada	2024-01-02 12:13:00	2025-09-10 16:38:00	CAD	...	
31498	ZR_28483_CAND	Female	Yes	No	Yes	Scarborough	Canada	2024-01-02 12:13:00	2026-02-18 14:46:00	CAD	...	
31499	ZR_28482_CAND	Female	Yes	No	Yes	AYR	Canada	2024-01-02	2025-03-10	CAD	...	

	Candidate ID.1	Gender	Email	Phone Number	Mobile Number	City	Country	Created Time	Modified Time	Currency	...	Number_En
								12:13:00	16:30:00			

31500 rows × 56 columns

In [18]:
```
categorical_columns = df.select_dtypes(include='object').columns

for col in categorical_columns:
    df[col] = df[col].fillna('Unknown')
```

In [19]:
```
df.isnull().sum()
```

Out[19]:

Candidate ID.1	0
Gender	0
Email	0
Phone Number	0
Mobile Number	0
City	0
Country	0
Created Time	0
Modified Time	0
Currency	0
Last Activity Time	0
Last emailed	0
Source	0
Email Opt Out	0
Is Hot Candidate	0
Is Locked	0
Candidate Status	0
Initial Contact Date	7541
Institute 1 Sign Up Date	0
Hire Date	29784
Ethnicity	0
Religion	0
Position Category	0
License Class	0
License year	0
Years of Experience	0
Current and Previous Mortgage	0
Equifax Credit Score	0
Criminal History	0
Prior Discipline Action by a Regulator	0
Professional License Suspension	0
Candidate Stage	0
Realtor License	0
Hiring Validation	0
1 Month Fee Waiver	0
Work Type	0
Licensing Status	0
Province	0
Parked	0
Pre-Interview Info Email	0
Post Interview Info Email	0
Discussion Type	0

```
Live Transfer                              0
Discussion Stage                           0
Schedule Discussion                        0
Email_Encoded                              0
Phone Number_Encoded                       0
Mobile Number_Encoded                      0
Email Opt Out_Encoded                      0
Is Locked_Encoded                          0
Equifax Credit Score_Encoded               0
Hiring Validation_Encoded                  0
Schedule Discussion_Encoded                0
Parked_Encoded                             0
Pre-Interview Info Email_Encoded           0
Post Interview Info Email_Encoded          0
dtype: int64
```

In [20]:
```
df.shape
```

Out[20]:
```
(31500, 56)
```

In [21]:
```
df.duplicated().sum()
```

Out[21]:
```
np.int64(0)
```

In [22]:
```
#To Trim and Standardize the Dataset
df.select_dtypes(include='object').columns
```

Out[22]:
```
Index(['Candidate ID.1', 'Gender', 'Email', 'Phone Number', 'Mobile Number',
       'City', 'Country', 'Currency', 'Last Activity Time', 'Last emailed',
       'Source', 'Email Opt Out', 'Is Hot Candidate', 'Is Locked',
       'Candidate Status', 'Institute 1 Sign Up Date', 'Ethnicity', 'Religion',
       'Position Category', 'License Class', 'Current and Previous Mortgage',
       'Equifax Credit Score', 'Criminal History',
       'Prior Discipline Action by a Regulator',
       'Professional License Suspension', 'Candidate Stage', 'Realtor License',
       'Hiring Validation', '1 Month Fee Waiver', 'Work Type',
       'Licensing Status', 'Province', 'Parked', 'Pre-Interview Info Email',
       'Post Interview Info Email', 'Discussion Type', 'Live Transfer',
       'Discussion Stage', 'Schedule Discussion'],
      dtype='str')
```

In [23]:
```
for col in df.select_dtypes(include='object').columns:
    print(f"\n{col}")
    print(df[col].unique())
```

```
Candidate ID.1
<ArrowStringArray>
['ZR_28480_CAND', 'ZR_28476_CAND', 'ZR_28475_CAND', 'ZR_28474_CAND',
 'ZR_28473_CAND', 'ZR_28472_CAND', 'ZR_28471_CAND', 'ZR_28470_CAND',
 'ZR_28469_CAND', 'ZR_28468_CAND',
 ...
 'ZR_28492_CAND', 'ZR_28491_CAND', 'ZR_28490_CAND', 'ZR_28489_CAND',
 'ZR_28487_CAND', 'ZR_28486_CAND', 'ZR_28485_CAND', 'ZR_28484_CAND',
 'ZR_28483_CAND', 'ZR_28482_CAND']
Length: 31500, dtype: str

Gender
<ArrowStringArray>
['Male', 'Female']
Length: 2, dtype: str

Email
<ArrowStringArray>
['Yes', 'No']
Length: 2, dtype: str

Phone Number
<ArrowStringArray>
['No', 'Yes']
Length: 2, dtype: str

Mobile Number
<ArrowStringArray>
['Yes', 'No']
Length: 2, dtype: str

City
<ArrowStringArray>
[
'NORTH YORK',

'Cambridge',

'Brampton',

'Toronto',
```

```
'Oshawa',

'Hamilton',

'Ancaster',

'Innisfil',

'AJAX',

'Unknown',
 ...

'newmarket',

'Hamilton, ON',

'Drumbo',

'Clarence Creek',

'Delta',

'Cambridge ON',

'Cookstown',

'Guleph',
 'Hamilton (West Kentley / McQuesten / Parkview / Hamilton Beach / East Industrial Sector / Normanhurst / Homeside /
East Crown Point)',

'Brampton ,ON']
Length: 905, dtype: str

Country
<ArrowStringArray>
['Canada']
Length: 1, dtype: str

Currency
<ArrowStringArray>
['CAD', 'PKR', 'INR']
```

```
Length: 3, dtype: str

Last Activity Time
<ArrowStringArray>
[  '3/6/2026 13:01',   '5/6/2026 14:43',  '2/10/2026 15:40',
  '4/15/2026 17:30',  '3/13/2026 12:00',  '3/30/2026 11:36',
 '11/14/2025 12:59',    '1/9/2026 9:59',  '9/23/2025 12:54',
   '5/9/2025 11:55',
 ...
  '3/16/2026 16:57',   '2/4/2026 12:33',  '4/27/2026 17:12',
  '3/25/2026 16:15', '10/20/2025 12:33',  '3/18/2026 10:50',
  '4/13/2026 17:51',  '4/29/2026 11:03',  '3/21/2025 16:16',
  '3/31/2026 15:41']
Length: 14003, dtype: str

Last emailed
<ArrowStringArray>
[         'Unknown',   '1/5/2024 13:35',   '1/9/2024 13:02',
  '11/5/2024 15:19', '12/21/2023 14:30', '12/20/2023 16:09',
 '12/20/2023 16:02',   '5/2/2024 15:33', '12/20/2023 13:09',
  '7/18/2024 14:23',
 ...
  '1/10/2024 11:13',   '7/8/2025 15:37',  '9/25/2024 16:22',
  '1/16/2024 15:33',  '1/17/2024 14:35',   '5/8/2024 15:39',
  '1/22/2024 12:52',  '1/14/2025 12:24',   '1/2/2024 14:00',
   '1/18/2024 9:06']
Length: 10242, dtype: str

Source
<ArrowStringArray>
[       'Referral',     'Institute 1',         'Call-In',         'Emailed',
         'Website',  'Social Media 1',              'BI',             'MCD',
     'Institute 3',       'Cold Call',     'Institute 2',  'Existing Agent',
 'Search Engine 2',    'Resume Inbox',    'Market Place',        'Job Site',
  'Social Media 2',      'CareerSite',         'Walk-In',      'Trade Show',
      'Emailed-In', 'Job Site Resume',    'Social Media',      'Mass Email',
     'Institute 4',          'Unkown']
Length: 26, dtype: str

Email Opt Out
<ArrowStringArray>
['No', 'Yes']
```

```
Length: 2, dtype: str

Is Hot Candidate
<ArrowStringArray>
['No', 'Yes']
Length: 2, dtype: str

Is Locked
<ArrowStringArray>
['Yes', 'No']
Length: 2, dtype: str

Candidate Status
<ArrowStringArray>
[                     'Hired',   'Joined Another Brokerage',
                    'No-Show',        'Interview-Scheduled',
 'Rejected by hiring manager',       'Attempted to Contact',
       'Following Up to Join',                   'On-Hold',
            'Junk candidate',             'Non Responsive',
  'Candidate Refused to Meet',  'Candidate Refused to Join',
                'Associated',  'Interview-to-be-Scheduled',
               'Unqualified',                'Pre-Course',
                       'New',                    'Joined',
           'Offer-Withdrawn',                'Offer-Made']
Length: 20, dtype: str

Institute 1 Sign Up Date
<ArrowStringArray>
[   'Unknown', '12/19/2023', '12/15/2023', '12/16/2023', '12/18/2023',
 '12/14/2023', '12/12/2023', '12/13/2023', '12/11/2023',  '12/8/2023',
 ...
 '12/29/2023', '12/30/2023', '12/31/2023',   '1/1/2024', '12/20/2023',
 '12/21/2023', '12/22/2023', '12/23/2023', '12/24/2023', '12/25/2023']
Length: 1677, dtype: str

Ethnicity
<ArrowStringArray>
[     'South Asian',        'Caucasian',          'African',
   'Middle Eastern',            'Black', 'South-East Asian',
   'Latin American',         'Hispanic',       'East Asian',
        'Caribbean',       'Indigenous',          'Unknown',
         'European', 'Pacific Islander']
```

```
Length: 14, dtype: str

Religion
<ArrowStringArray>
[               'Hinduism',            'Christianity',
                 'Sikhism',                   'Islam',
                'Buddhism',                'Unknown',
                 'Judaism',                'Atheist',
                   'Unkno', 'Christianity (Orthodox)',
                 'Jainism',          'Zoroastrianism']
Length: 12, dtype: str

Position Category
<ArrowStringArray>
[        'Mortgage Agent',       'Principal Broker',                 'Broker',
                  'Agent',  'Mortgage Professional', 'Referral Partner (RPP)',
                'Realtor']
Length: 7, dtype: str

License Class
<ArrowStringArray>
[         'Unknown',    'Agent Level 1',    'Agent Level 2',
            'Agent',           'Broker', 'Principal Broker']
Length: 6, dtype: str

Current and Previous Mortgage
<ArrowStringArray>
[
'X',

'Unknown',

'Affinity Mortgage Solutions Inc.',

'Mountainview Mortgage Inc.',

'AKAL Mortgages Inc.',

'MORTGAGE ALLIANCE',

'BRX Mortgage Inc.',
```

```
'Mortgage Alliance Company of Canada Inc.',

'8Twelve Mortgage Corporation',

'Better Mortgage',
 ...
 'Dominion Lending Centres
""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""
""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""
""""""""""""""""""""""Mortgages Are Marvellou
s""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""
""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""""
""""""""""""""""""""""""',

'12060485 CANADA INC. operating as WOODBINE MORTGAGE CENTRE',

'Kingsdale Mortgage',

'Dominion Lending Centres ""Mortgages Are Marvellous""',

'Forest City Funding',

'Mortgage Bridge Canada',

'Landeal Capital Inc',

'Mortgage Outlet',

'Promise Mortgage and Financial',

'Mortgages On Point']
Length: 1990, dtype: str

Equifax Credit Score
<ArrowStringArray>
['Yes', 'No', 'Unknown']
Length: 3, dtype: str

Criminal History
<ArrowStringArray>
[                              'Not Applicable',
                                      'Unknown',
```

```
               'Highway Traffic Act Violation',
                        'Criminal Background',
                    'Driving Under Influence',
          'Criminal Background;Not Applicable',
 'Driving Under Influence;Criminal Background']
Length: 7, dtype: str

Prior Discipline Action by a Regulator
<ArrowStringArray>
['Unknown', 'Yes']
Length: 2, dtype: str

Professional License Suspension
<ArrowStringArray>
['Not Applicable', 'Unknown']
Length: 2, dtype: str

Candidate Stage
<ArrowStringArray>
['In Review', 'New', 'Unknown', 'Hired', 'Engaged', 'Available']
Length: 6, dtype: str

Realtor License
<ArrowStringArray>
['No', 'Unknown', 'Yes']
Length: 3, dtype: str

Hiring Validation
<ArrowStringArray>
['Yes', 'No']
Length: 2, dtype: str

1 Month Fee Waiver
<ArrowStringArray>
['No', 'Unknown', 'Yes', 'Not Applicable']
Length: 4, dtype: str

Work Type
<ArrowStringArray>
['Unknown', 'Full Time', 'Part Time']
Length: 3, dtype: str
```

```
Licensing Status
<ArrowStringArray>
[    'Authorized to Sell',               'Expired',               'Unknown',
 'Not Authorized to Sell',           'Surrendered',               'Pending',
               'Revoked',             'Suspended',    'Application Refused',
            'Non-Active']
Length: 10, dtype: str

Province
<ArrowStringArray>
[                  'Ontario',                  'Unknown',
                   'Alberta',                       'ON',
             'New Brunswick',          'British Columbia',
              'Saskatchewan',                  'Manitoba',
               'Nova Scotia',      'Prince Edward Island',
 'NewFoundLand and Labrador',                   'Quebec',
                    'Alaska',                  'Florida',
                  'Michigan']
Length: 15, dtype: str

Parked
<ArrowStringArray>
['No', 'Unknown', 'Yes']
Length: 3, dtype: str

Pre-Interview Info Email
<ArrowStringArray>
['No', 'Yes', 'Unknown']
Length: 3, dtype: str

Post Interview Info Email
<ArrowStringArray>
['No', 'Yes', 'Unknown']
Length: 3, dtype: str

Discussion Type
<ArrowStringArray>
['Unknown', 'Scheduled', 'Direct Transfer']
Length: 3, dtype: str

Live Transfer
<ArrowStringArray>
```

```
['Unknown', 'Offered', 'Not Offered']
Length: 3, dtype: str

Discussion Stage
<ArrowStringArray>
['Unknown', 'Stage 1', 'Stage 2']
Length: 3, dtype: str

Schedule Discussion
<ArrowStringArray>
['No', 'Yes', 'Unknown']
Length: 3, dtype: str
```

In [24]:
```python
categorical_columns = df.select_dtypes(include='object').columns

for col in categorical_columns:
    df[col] = df[col].astype(str).str.strip().str.title()
```

In [25]:
```python
#Checking the Result
for col in categorical_columns:
    print(f"\n{col}")
    print(df[col].value_counts())
```

```
Candidate ID.1
Candidate ID.1
Zr_28480_Cand    1
Zr_28476_Cand    1
Zr_28475_Cand    1
Zr_28474_Cand    1
Zr_28473_Cand    1
                ..
Zr_28486_Cand    1
Zr_28485_Cand    1
Zr_28484_Cand    1
Zr_28483_Cand    1
Zr_28482_Cand    1
Name: count, Length: 31500, dtype: int64

Gender
Gender
Male      21994
Female     9506
Name: count, dtype: int64

Email
Email
Yes    25219
No      6281
Name: count, dtype: int64

Phone Number
Phone Number
No     27216
Yes     4284
Name: count, dtype: int64

Mobile Number
Mobile Number
Yes    27843
No      3657
Name: count, dtype: int64

City
City
Unknown
```

```
22139
Brampton
948
Toronto
873
Calgary
708
Mississauga
629

...
Clarence Creek
1
Cambridge On
1
Guleph
1
Hamilton (West Kentley / Mcquesten / Parkview / Hamilton Beach / East Industrial Sector / Normanhurst / Homeside / Ea
st Crown Point)        1
Brampton ,On
1
Name: count, Length: 680, dtype: int64

Country
Country
Canada    31500
Name: count, dtype: int64

Currency
Currency
Cad    31470
Pkr       19
Inr       11
Name: count, dtype: int64

Last Activity Time
Last Activity Time
4/30/2026 13:50     1801
3/7/2025 8:59       1445
5/6/2026 14:40       200
5/6/2026 14:43       199
5/6/2026 14:55       199
```

```
                    ...
10/20/2025 12:33       1
3/18/2026 10:50        1
4/29/2026 11:03        1
3/21/2025 16:16        1
3/31/2026 15:41        1
Name: count, Length: 14003, dtype: int64

Last emailed
Last emailed
Unknown             20272
2/21/2025 14:23         9
10/31/2023 9:49         9
11/21/2023 11:40        7
11/3/2023 9:14          7
                    ...
5/8/2024 15:39          1
1/22/2024 12:52         1
1/14/2025 12:24         1
1/2/2024 14:00          1
1/18/2024 9:06          1
Name: count, Length: 10242, dtype: int64

Source
Source
Institute 1        9883
Website            9839
Bi                 7299
Institute 3        1311
Mcd                 601
Institute 2         559
Cold Call           554
Institute 4         469
Referral            330
Call-In             173
Emailed             106
Unkown               80
Job Site Resume      70
Job Site             55
Market Place         44
Careersite           44
Social Media 2       41
```

```
Existing Agent       11
Social Media 1        8
Walk-In               8
Search Engine 2       6
Emailed-In            4
Trade Show            2
Resume Inbox          1
Social Media          1
Mass Email            1
Name: count, dtype: int64

Email Opt Out
Email Opt Out
No     26450
Yes     5050
Name: count, dtype: int64

Is Hot Candidate
Is Hot Candidate
No     31493
Yes        7
Name: count, dtype: int64

Is Locked
Is Locked
No     29985
Yes     1515
Name: count, dtype: int64

Candidate Status
Candidate Status
Associated                   9933
New                          7596
Attempted To Contact         4025
Hired                        1796
Interview-Scheduled          1649
Candidate Refused To Join    1275
Following Up To Join          983
On-Hold                       838
No-Show                       788
Joined Another Brokerage      709
Junk Candidate                670
```

```
Non Responsive                607
Candidate Refused To Meet     455
Interview-To-Be-Scheduled      53
Rejected By Hiring Manager     50
Joined                         38
Offer-Withdrawn                24
Pre-Course                      6
Offer-Made                      3
Unqualified                     2
Name: count, dtype: int64

Institute 1 Sign Up Date
Institute 1 Sign Up Date
Unknown       26042
2/7/2020         21
12/17/2019       21
2/19/2020        20
1/15/2020        18
               ...
12/31/2023        1
1/1/2024          1
12/22/2023        1
12/24/2023        1
12/25/2023        1
Name: count, Length: 1677, dtype: int64

Ethnicity
Ethnicity
Caucasian          11752
South Asian        10901
Middle Eastern      2755
East Asian          2465
Hispanic             926
Black                800
African              676
South-East Asian     539
Unknown              236
Latin American       211
Caribbean            154
European              70
Pacific Islander      10
Indigenous             5
```

```
Name: count, dtype: int64

Religion
Religion
Christianity               15343
Hinduism                    5695
Islam                       3945
Sikhism                     3102
Buddhism                    1579
Unknown                     1188
Judaism                      393
Atheist                      248
Jainism                        3
Zoroastrianism                 2
Unkno                          1
Christianity (Orthodox)        1
Name: count, dtype: int64

Position Category
Position Category
Mortgage Agent             31366
Realtor                       61
Agent                         46
Broker                        13
Principal Broker               7
Referral Partner (Rpp)         6
Mortgage Professional          1
Name: count, dtype: int64

License Class
License Class
Unknown              13872
Agent Level 1         9432
Agent Level 2         4523
Agent                 1462
Broker                1451
Principal Broker       760
Name: count, dtype: int64

Current and Previous Mortgage
Current and Previous Mortgage
Unknown                                17807
```

```
X                                          782
Mortgage Alliance Company Of Canada Inc.   640
Ma Mortgage Architects Inc.                335
Real Mortgage Associates Inc.              285
                                          ...
Forest City Funding                          1
Mortgage Bridge Canada                       1
Landeal Capital Inc                          1
Promise Mortgage And Financial               1
Mortgages On Point                           1
Name: count, Length: 1917, dtype: int64

Equifax Credit Score
Equifax Credit Score
No         29568
Yes         1930
Unknown        2
Name: count, dtype: int64

Criminal History
Criminal History
Unknown                                    30099
Not Applicable                              1373
Highway Traffic Act Violation                 12
Criminal Background                            9
Driving Under Influence                        5
Criminal Background;Not Applicable             1
Driving Under Influence;Criminal Background    1
Name: count, dtype: int64

Prior Discipline Action by a Regulator
Prior Discipline Action by a Regulator
Unknown    31492
Yes            8
Name: count, dtype: int64

Professional License Suspension
Professional License Suspension
Unknown           29992
Not Applicable     1508
Name: count, dtype: int64
```

```
Candidate Stage
Candidate Stage
New          18356
Engaged       4912
In Review     4537
Unknown       3586
Hired           85
Available       24
Name: count, dtype: int64

Realtor License
Realtor License
Unknown    30567
No           820
Yes          113
Name: count, dtype: int64

Hiring Validation
Hiring Validation
No     30042
Yes     1458
Name: count, dtype: int64

1 Month Fee Waiver
1 Month Fee Waiver
Unknown           30080
No                  578
Yes                 528
Not Applicable      314
Name: count, dtype: int64

Work Type
Work Type
Unknown      30309
Part Time      631
Full Time      560
Name: count, dtype: int64

Licensing Status
Licensing Status
Authorized To Sell        13392
Unknown                   12215
```

```
Expired                    4406
Not Authorized To Sell     1121
Surrendered                 248
Pending                      65
Suspended                    38
Revoked                      13
Application Refused           1
Non-Active                    1
Name: count, dtype: int64

Province
Province
Ontario                    27355
On                          1653
Alberta                     1524
Quebec                       368
Unknown                      363
British Columbia             189
New Brunswick                 18
Saskatchewan                  10
Manitoba                       7
Nova Scotia                    5
Newfoundland And Labrador      3
Prince Edward Island           2
Alaska                         1
Florida                        1
Michigan                       1
Name: count, dtype: int64

Parked
Parked
No         30011
Yes         1408
Unknown       81
Name: count, dtype: int64

Pre-Interview Info Email
Pre-Interview Info Email
No         26310
Yes         5109
Unknown       81
Name: count, dtype: int64
```

```
Post Interview Info Email
Post Interview Info Email
No         30766
Yes          653
Unknown       81
Name: count, dtype: int64

Discussion Type
Discussion Type
Unknown            29510
Direct Transfer     1597
Scheduled            393
Name: count, dtype: int64

Live Transfer
Live Transfer
Unknown        29511
Offered         1984
Not Offered        5
Name: count, dtype: int64

Discussion Stage
Discussion Stage
Unknown    29512
Stage 1     1982
Stage 2        6
Name: count, dtype: int64

Schedule Discussion
Schedule Discussion
No         29437
Yes         1982
Unknown       81
Name: count, dtype: int64
```

Following the data preparation process, the final analytical dataset consisted of 31,500 unique candidate records and 56 variables, with no duplicate records identified. The preparation involved removing rows without a valid candidate identifier, standardising inconsistent formatting, converting relevant date fields into a consistent datetime format, and encoding Yes/No variables for subsequent analysis. Missing values were also assessed and treated according to the nature of each variable, with categorical variables assigned an “Unknown” category and numerical variables handled using median imputation where appropriate. This

process ensured that the dataset was structured and sufficiently consistent for the subsequent descriptive, exploratory, recruitment funnel, and predictive analyses.

Descriptive Analysis

In [26]:
```
# Total Count
len(df)
```

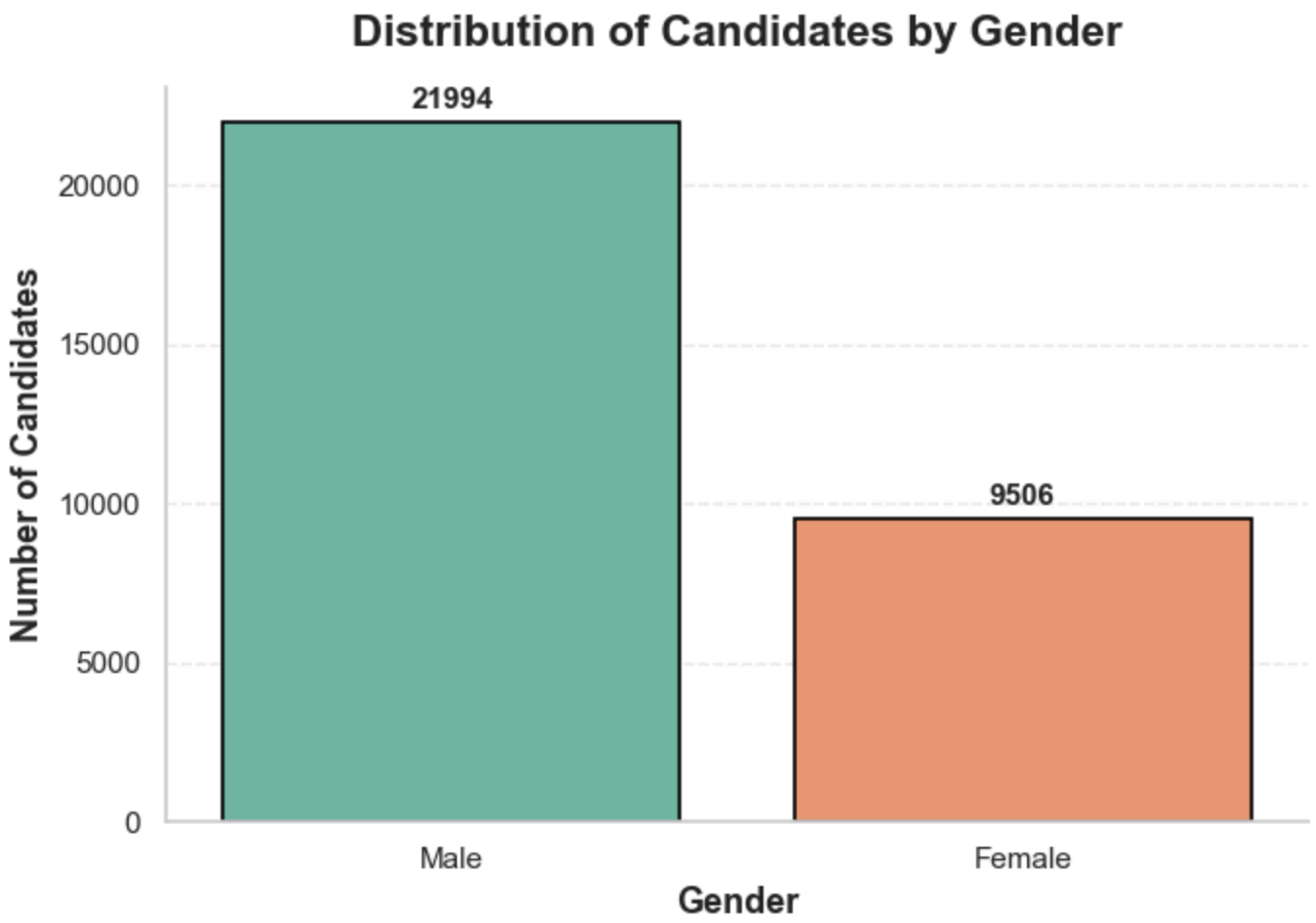
Out[26]:
```
31500
```

In [27]:
```
#Gender Count
df['Gender'].value_counts()
```

Out[27]:
```
Gender
Male      21994
Female     9506
Name: count, dtype: int64
```

In [28]:
```
#Bar Chart
sns.set_theme(style="whitegrid")

plt.figure(figsize=(7,5))

ax = sns.countplot(
    data=df,
    x='Gender',
    hue='Gender',
    palette='Set2',
    legend=False,
    edgecolor='black',
    linewidth=1.2)

# Title and axis labels
plt.title('Distribution of Candidates by Gender',
          fontsize=16,
          fontweight='bold',
          pad=15)

plt.xlabel('Gender', fontsize=13, fontweight='bold')
```

```
plt.ylabel('Number of Candidates', fontsize=13, fontweight='bold')

# Add value labels
for container in ax.containers:
    ax.bar_label(container,
                 fontsize=11,
                 fontweight='bold',
                 padding=3)

# Improve appearance
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.xticks(fontsize=11)
plt.yticks(fontsize=11)

plt.grid(axis='y', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()
```

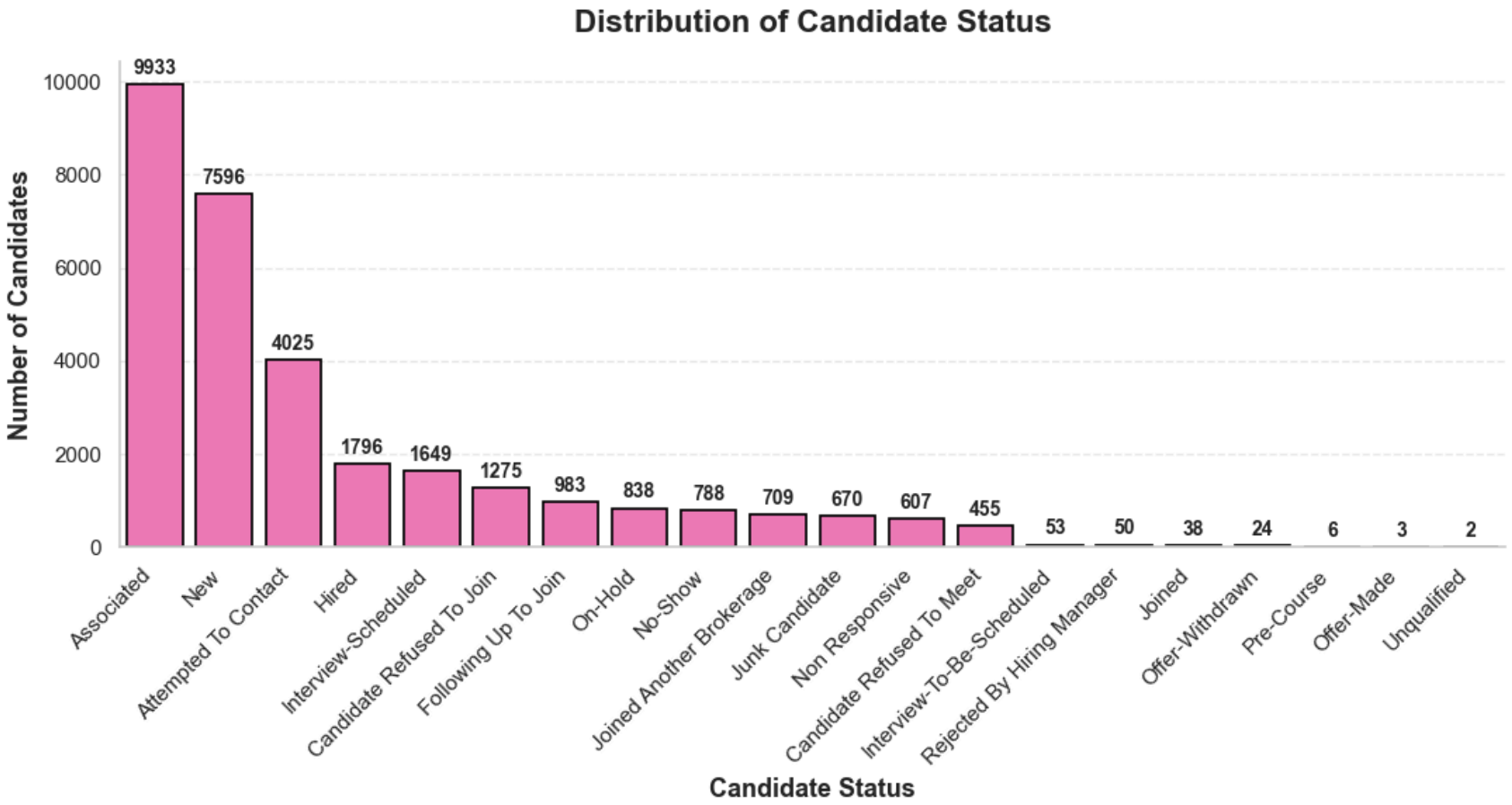


In [29]:
```
#Candidate Status
df['Candidate Status'].value_counts()
```

Out[29]:
```
Candidate Status
Associated                    9933
New                           7596
Attempted To Contact          4025
Hired                         1796
Interview-Scheduled           1649
Candidate Refused To Join     1275
Following Up To Join           983
On-Hold                        838
No-Show                        788
Joined Another Brokerage       709
Junk Candidate                 670
Non Responsive                 607
Candidate Refused To Meet      455
Interview-To-Be-Scheduled       53
Rejected By Hiring Manager      50
Joined                          38
Offer-Withdrawn                 24
Pre-Course                       6
Offer-Made                       3
Unqualified                      2
Name: count, dtype: int64
```

In [30]:
```python
# Bar Chart for Candidate Status
candidate_status = df['Candidate Status'].value_counts()

# Professional style
sns.set_theme(style="whitegrid")

plt.figure(figsize=(11,6))

ax = sns.barplot(
    x=candidate_status.index,
    y=candidate_status.values,
    color='hotpink',
    edgecolor='black',
    linewidth=1.2
)

# Titles
plt.title('Distribution of Candidate Status',
          fontsize=16,
```

```
          fontweight='bold',
          pad=15)

plt.xlabel('Candidate Status',
           fontsize=13,
           fontweight='bold')

plt.ylabel('Number of Candidates',
           fontsize=13,
           fontweight='bold')

# Rotate labels
plt.xticks(rotation=45,
           ha='right',
           fontsize=11)

plt.yticks(fontsize=11)

# Add values on bars
for container in ax.containers:
    ax.bar_label(container,
                 fontsize=10,
                 fontweight='bold',
                 padding=3)

# Improve appearance
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.grid(axis='y', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()
```



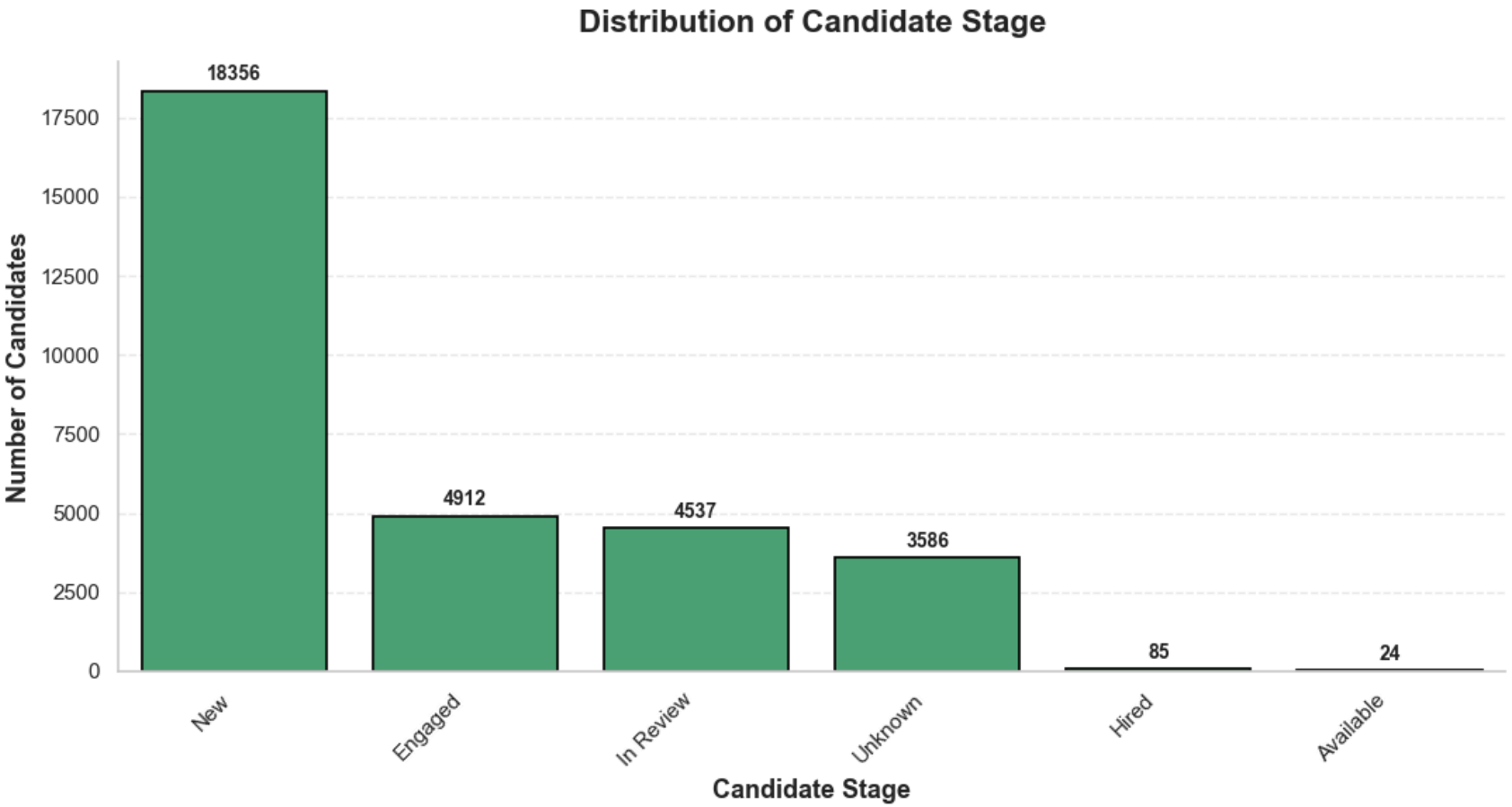
In [31]:
```
#Candidate Stage
df['Candidate Stage'].value_counts()
```

Out[31]:
```
Candidate Stage
New          18356
Engaged       4912
In Review     4537
Unknown       3586
Hired           85
Available       24
Name: count, dtype: int64
```

In [32]:
```
# Bar Chart for Candidate Stage
candidate_stage = df['Candidate Stage'].value_counts()
```

```
# Professional style
sns.set_theme(style="whitegrid")

plt.figure(figsize=(11,6))

ax = sns.barplot(
    x=candidate_stage.index,
    y=candidate_stage.values,
    color='mediumseagreen',
    edgecolor='black',
    linewidth=1.2
)

# Title
plt.title('Distribution of Candidate Stage',
          fontsize=16,
          fontweight='bold',
          pad=15)

# Axis labels
plt.xlabel('Candidate Stage',
           fontsize=13,
           fontweight='bold')

plt.ylabel('Number of Candidates',
           fontsize=13,
           fontweight='bold')

# Rotate x-axis labels
plt.xticks(rotation=45,
           ha='right',
           fontsize=11)

plt.yticks(fontsize=11)

# Add values on top of bars
for container in ax.containers:
    ax.bar_label(container,
                 fontsize=10,
                 fontweight='bold',
                 padding=3)
```

```
# Remove unnecessary borders
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Light horizontal grid
plt.grid(axis='y', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()
```

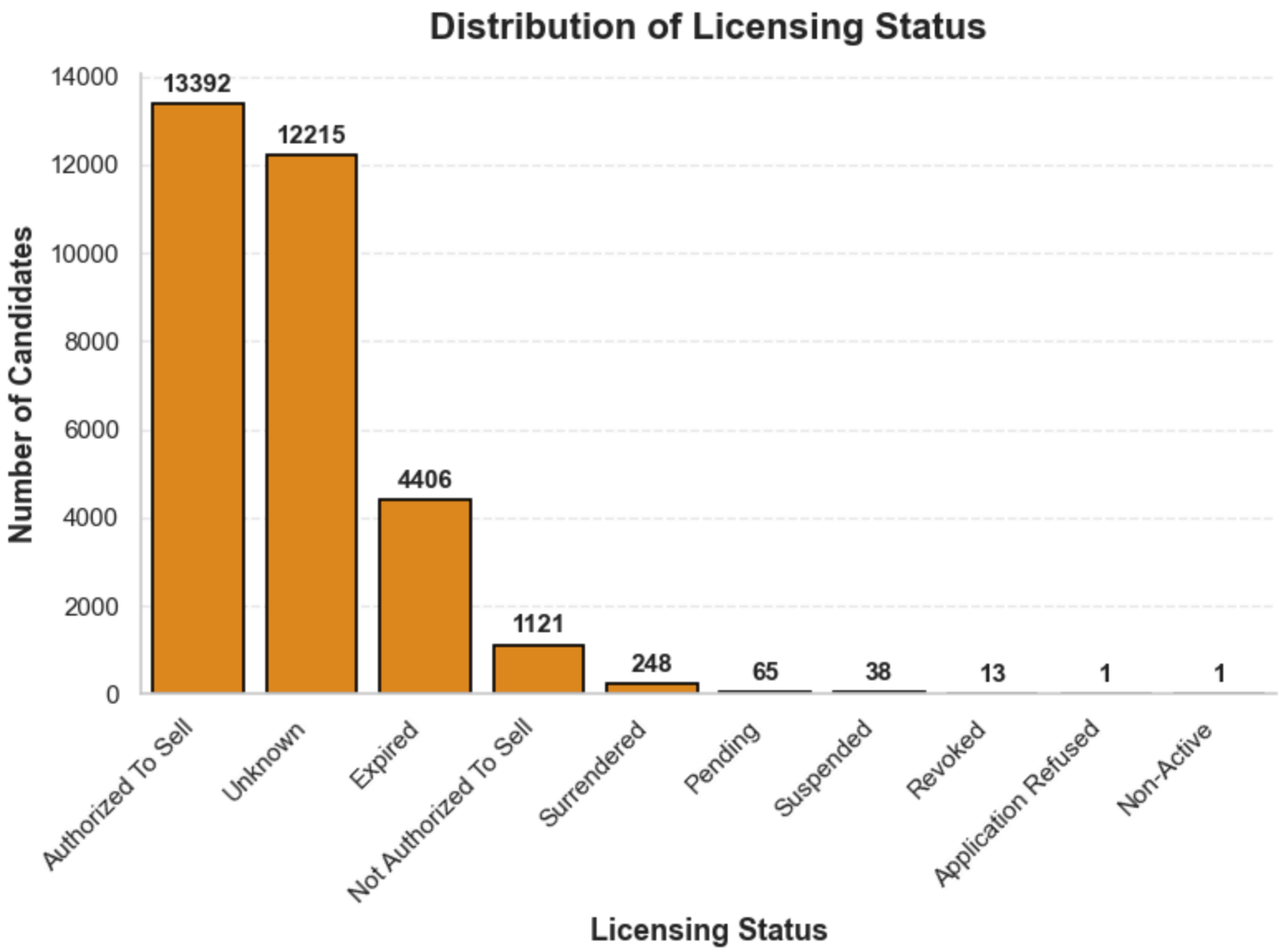


In [33]:
```
#Licensing Status
df['Licensing Status'].value_counts()
```

Out[33]:
```
Licensing Status
Authorized To Sell        13392
Unknown                   12215
Expired                    4406
Not Authorized To Sell     1121
Surrendered                 248
Pending                      65
Suspended                    38
Revoked                      13
Application Refused           1
Non-Active                    1
Name: count, dtype: int64
```

In [34]:
```
#Bar Chart for Licensing Status
licensing_status = df['Licensing Status'].value_counts()

# Professional style
sns.set_theme(style="whitegrid")

plt.figure(figsize=(8,6))

ax = sns.barplot(
    x=licensing_status.index,
    y=licensing_status.values,
    color='darkorange',
    edgecolor='black',
    linewidth=1.2
)

# Title
plt.title('Distribution of Licensing Status',
          fontsize=16,
          fontweight='bold',
          pad=15)

# Axis labels
plt.xlabel('Licensing Status',
           fontsize=13,
           fontweight='bold')

plt.ylabel('Number of Candidates',
           fontsize=13,
```

```
            fontweight='bold')

# Rotate x-axis labels
plt.xticks(rotation=45,
           ha='right',
           fontsize=11)

# Tick formatting
plt.xticks(fontsize=11)
plt.yticks(fontsize=11)

# Add value labels
for container in ax.containers:
    ax.bar_label(container,
                 fontsize=11,
                 fontweight='bold',
                 padding=3)

# Remove unnecessary borders
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Add light grid
plt.grid(axis='y', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()
```

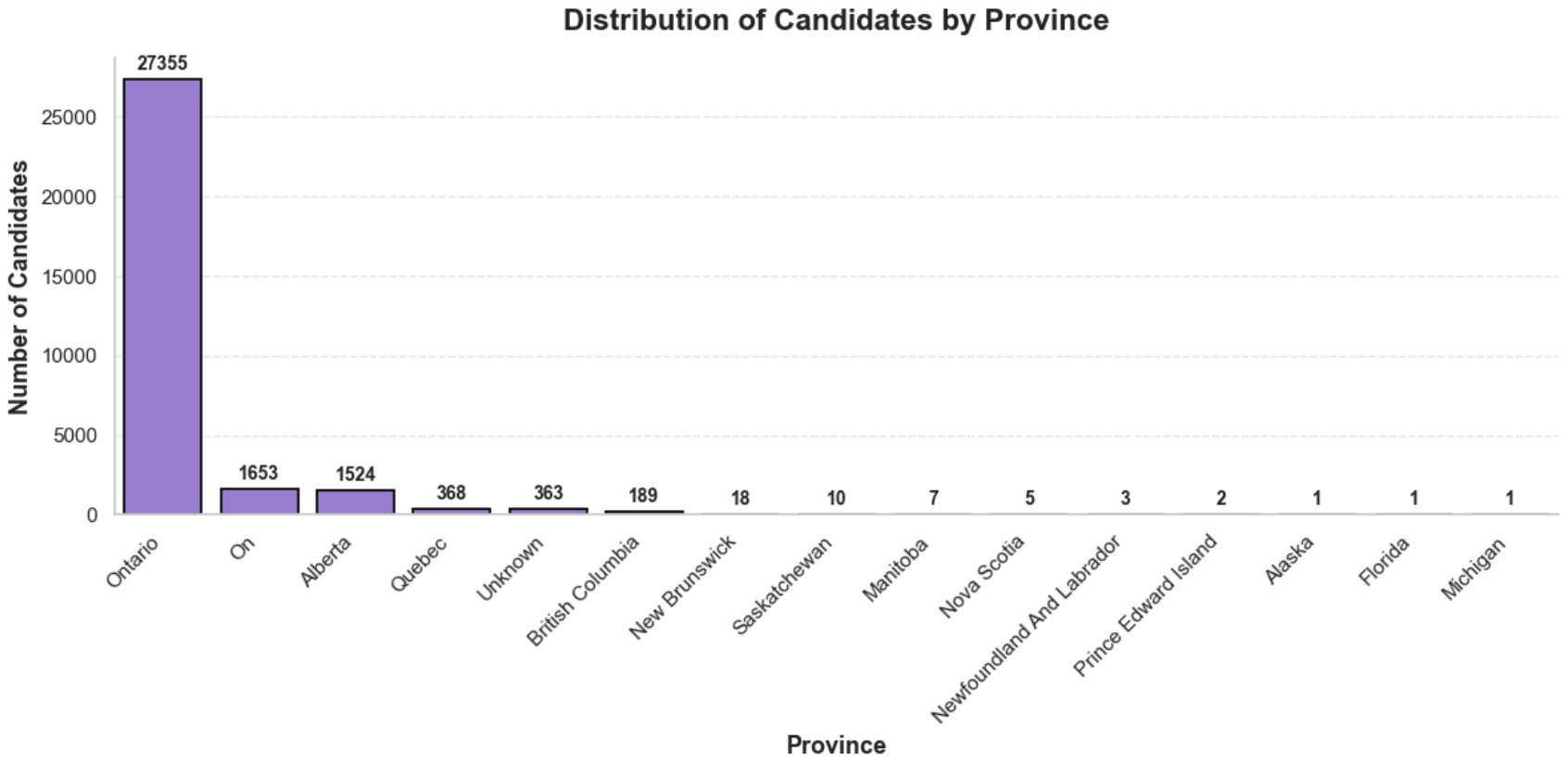


In [35]:
```
#Province
df['Province'].value_counts()
```

Out[35]:
```
Province
Ontario                      27355
On                            1653
Alberta                       1524
Quebec                         368
Unknown                        363
British Columbia               189
New Brunswick                   18
Saskatchewan                    10
Manitoba                         7
Nova Scotia                      5
Newfoundland And Labrador        3
Prince Edward Island             2
Alaska                           1
Florida                          1
Michigan                         1
Name: count, dtype: int64
```

In [36]:
```
#Bar Chart for Province
province = df['Province'].value_counts()

# Professional style
sns.set_theme(style="whitegrid")

plt.figure(figsize=(12,6))

ax = sns.barplot(
    x=province.index,
    y=province.values,
    color='mediumpurple',
    edgecolor='black',
    linewidth=1.2
)

# Title
plt.title('Distribution of Candidates by Province',
          fontsize=16,
          fontweight='bold',
          pad=15)

# Axis labels
plt.xlabel('Province',
```

```
            fontsize=13,
            fontweight='bold')

plt.ylabel('Number of Candidates',
            fontsize=13,
            fontweight='bold')

# Rotate province names
plt.xticks(rotation=45,
            ha='right',
            fontsize=11)

plt.yticks(fontsize=11)

# Add count labels
for container in ax.containers:
    ax.bar_label(container,
                 fontsize=10,
                 fontweight='bold',
                 padding=3)

# Remove unnecessary borders
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Add light grid
plt.grid(axis='y',
         linestyle='--',
         alpha=0.4)

plt.tight_layout()
plt.show()
```



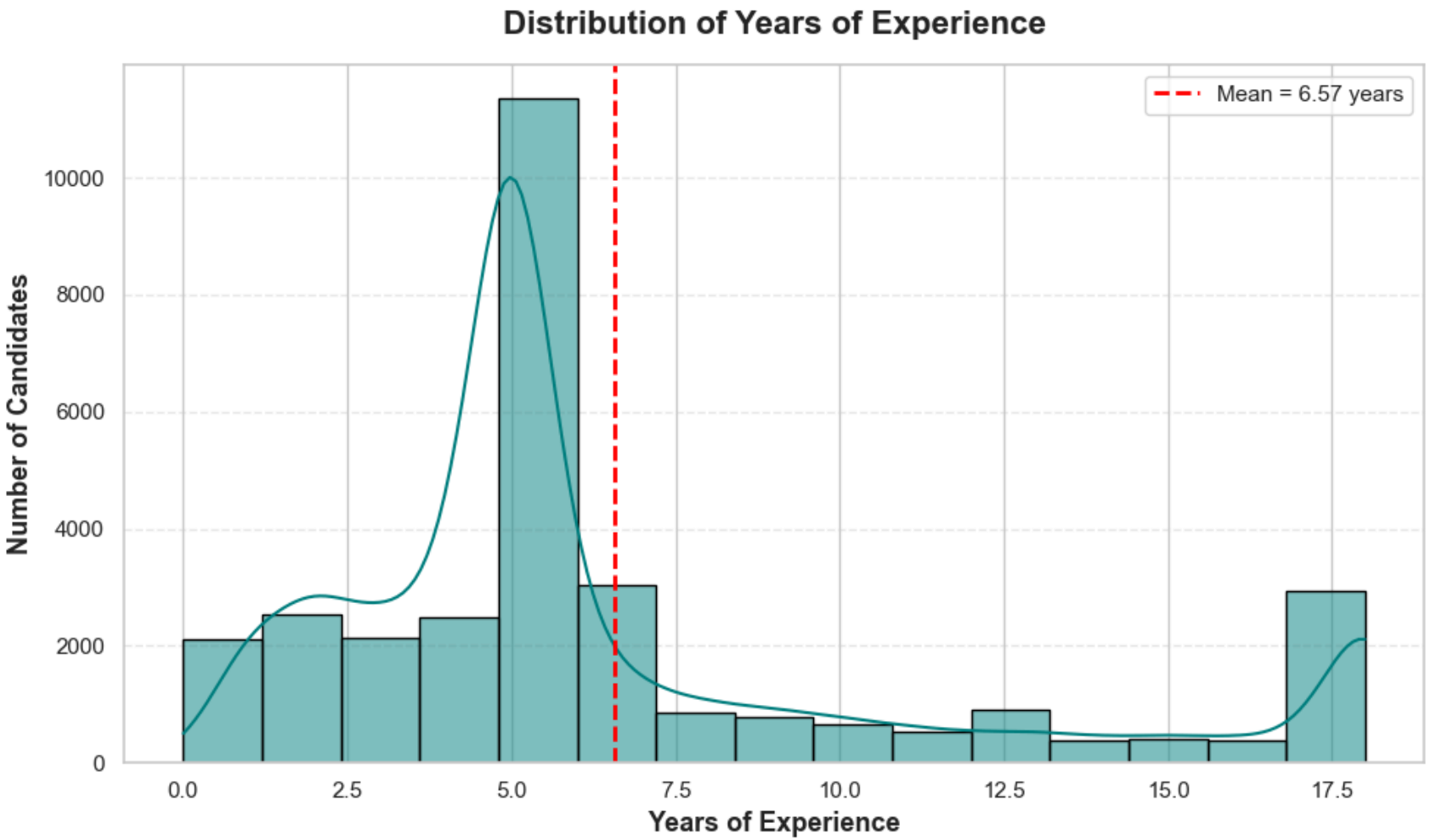
In [37]:
```
#Years of Experience
df['Years of Experience'].describe()
```

Out[37]:
```
count    31500.000000
mean         6.566032
std          4.733863
min          0.000000
25%          4.000000
50%          5.000000
75%          7.000000
max         18.000000
Name: Years of Experience, dtype: float64
```

In [38]:
```
# Histogram for Years of Experience
sns.set_theme(style="whitegrid")
```

```
plt.figure(figsize=(10,6))

ax = sns.histplot(
    data=df,
    x='Years of Experience',
    bins=15,
    color='teal',
    edgecolor='black',
    linewidth=1,
    kde=True
)

# Calculate the mean
mean_exp = df['Years of Experience'].mean()

# Add a vertical line for the mean
plt.axvline(mean_exp,
            color='red',
            linestyle='--',
            linewidth=2,
            label=f'Mean = {mean_exp:.2f} years')

# Title and labels
plt.title('Distribution of Years of Experience',
          fontsize=16,
          fontweight='bold',
          pad=15)

plt.xlabel('Years of Experience',
           fontsize=13,
           fontweight='bold')

plt.ylabel('Number of Candidates',
           fontsize=13,
           fontweight='bold')

# Improve appearance
plt.xticks(fontsize=11)
plt.yticks(fontsize=11)

plt.grid(axis='y', linestyle='--', alpha=0.4)
```

```
plt.legend()

plt.tight_layout()
plt.show()
```



The descriptive analysis provided an initial overview of the 31,500 candidate records, highlighting the distribution of candidate characteristics, recruitment sources, candidate statuses and hiring outcomes. The findings showed that candidate volume was concentrated across a small number of recruitment sources, while the overall proportion progressing to validated hiring remained relatively low. These results provide a baseline understanding of the recruitment population and outcomes, supporting the subsequent exploratory and recruitment funnel analyses by identifying key patterns and areas requiring further investigation.

Funnel Analysis

In [39]:
```
# Inspect the values in each relevant column

print(df['Schedule Discussion'].value_counts(dropna=False))
```

```
Schedule Discussion
No         29437
Yes         1982
Unknown       81
Name: count, dtype: int64
```

In [40]:
```
print(df['Discussion Stage'].value_counts(dropna=False))
```

```
Discussion Stage
Unknown    29512
Stage 1     1982
Stage 2        6
Name: count, dtype: int64
```

In [41]:
```
print(df['Candidate Status'].value_counts(dropna=False))
```

```
Candidate Status
Associated                    9933
New                           7596
Attempted To Contact          4025
Hired                         1796
Interview-Scheduled           1649
Candidate Refused To Join     1275
Following Up To Join           983
On-Hold                        838
No-Show                        788
Joined Another Brokerage       709
Junk Candidate                 670
Non Responsive                 607
Candidate Refused To Meet      455
Interview-To-Be-Scheduled       53
Rejected By Hiring Manager      50
Joined                          38
Offer-Withdrawn                 24
Pre-Course                       6
Offer-Made                       3
Unqualified                      2
Name: count, dtype: int64
```

In [42]:
```
print(df['Licensing Status'].value_counts(dropna=False))
```

```
Licensing Status
Authorized To Sell        13392
Unknown                   12215
Expired                    4406
Not Authorized To Sell     1121
Surrendered                 248
Pending                      65
Suspended                    38
Revoked                      13
Application Refused           1
Non-Active                    1
Name: count, dtype: int64
```

In [43]:
```
print(df['Hiring Validation'].value_counts(dropna=False))
```

```
Hiring Validation
No     30042
Yes     1458
Name: count, dtype: int64
```

In [44]:
```
print(df['Discussion Type'].value_counts(dropna=False))
```

```
Discussion Type
Unknown            29510
Direct Transfer     1597
Scheduled            393
Name: count, dtype: int64
```

In [45]:
```
print(df['Candidate Stage'].value_counts(dropna=False))
```

```
Candidate Stage
New          18356
Engaged       4912
In Review     4537
Unknown       3586
Hired           85
Available       24
Name: count, dtype: int64
```

In [46]:
```
# Funnel Table

data_frame = len(df)

contacted = df['Candidate Status'].isin(['Attempted To Contact',
                                         'Following Up To Join',
                                         'On-Hold',
                                         'Joined Another Brokerage',
                                         'Non Responsive',
                                         'Hired',
                                         'Candidate Refused To Join',
                                         'No-Show',
                                         'Candidate Refused To Meet',
                                         'Rejected By Hiring Manager'
                                        ]).sum()

discussion_scheduled = df['Candidate Status'].isin([
    'Following Up To Join',
    'On-Hold',
```

```
    'Joined Another Brokerage',
    'Non Responsive',
    'Hired',
    'Candidate Refused To Join',
    'No-Show',
    'Candidate Refused To Meet',
    'Rejected By Hiring Manager']).sum()

discussion_completed = df['Candidate Status'].isin(['Following Up To Join',
                                                    'On-Hold',
                                                    'Joined Another Brokerage',
                                                    'Non Responsive',
                                                    'Hired',
                                                    'Candidate Refused To Join',
                                                    'Rejected By Hiring Manager'
                                                   ]).sum()

follow_up = df['Candidate Status'].isin(['Following Up To Join',
                                         'Joined Another Brokerage',
                                         'Hired',
                                         'Candidate Refused To Join'
                                        ]).sum()

validated = (df['Hiring Validation'] == 'Yes').sum()

hired = (df['Candidate Status'] == 'Hired').sum()
```

In [47]:

```
funnel = {
    'Stage':[
        'Data Frame',
        'Contacted',
        'Discussion Scheduled',
        'Discussion Completed',
        'Follow Up',
        'Hiring Validation',
        'Hired'
    ],
    'Candidates':[
        data_frame,
        contacted,
        discussion_scheduled,
        discussion_completed,
```

```
        follow_up,
        validated,
        hired
    ]
}

funnel_df = pd.DataFrame(funnel)

funnel_df
```

Out[47]:

	Stage	Candidates
0	Data Frame	31500
1	Contacted	11526
2	Discussion Scheduled	7501
3	Discussion Completed	6258
4	Follow Up	4763
5	Hiring Validation	1458
6	Hired	1796

In [48]:

```
# Calculate Conversion Rate
funnel_df['Conversion Rate (%)'] = (
    funnel_df['Candidates']
    / funnel_df['Candidates'].shift(1)
    *100
)

funnel_df.loc[0,'Conversion Rate (%)']=100

funnel_df
```

Out[48]:

	Stage	Candidates	Conversion Rate (%)
0	Data Frame	31500	100.000000
1	Contacted	11526	36.590476
2	Discussion Scheduled	7501	65.078952
3	Discussion Completed	6258	83.428876
4	Follow Up	4763	76.110578
5	Hiring Validation	1458	30.610959
6	Hired	1796	123.182442

In [49]:
```
# Calculate Drop Off Rate
funnel_df['Drop-off Rate (%)'] = (
    100 - funnel_df['Conversion Rate (%)']
)

funnel_df
```

Out[49]:

	Stage	Candidates	Conversion Rate (%)	Drop-off Rate (%)
0	Data Frame	31500	100.000000	0.000000
1	Contacted	11526	36.590476	63.409524
2	Discussion Scheduled	7501	65.078952	34.921048
3	Discussion Completed	6258	83.428876	16.571124
4	Follow Up	4763	76.110578	23.889422
5	Hiring Validation	1458	30.610959	69.389041
6	Hired	1796	123.182442	-23.182442

In [50]:
```
# Overall Completion Rate
completion_rate = (
    hired/data_frame
```

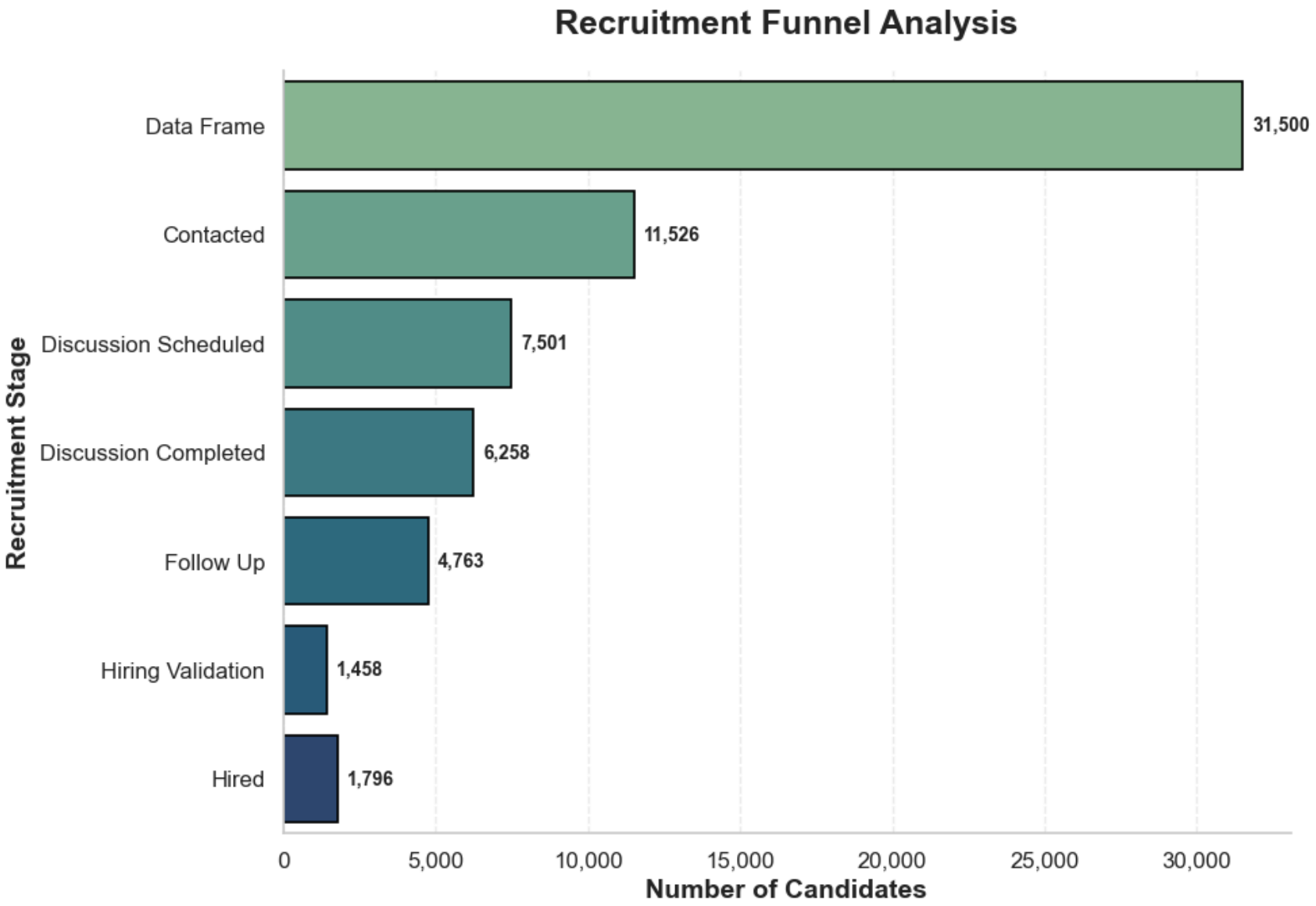
```
)*100

print(f"Overall Completion Rate: {completion_rate:.2f}%")
```

```
Overall Completion Rate: 5.70%
```

In [51]:

```
# Bar Chart

from matplotlib.ticker import FuncFormatter

sns.set_theme(style="whitegrid", font_scale=1.1)

plt.figure(figsize=(10,7))

colors = sns.color_palette("crest", len(funnel_df))

ax = sns.barplot(
    data=funnel_df,
    y='Stage',
    x='Candidates',
    palette=colors,
    edgecolor='black',
    linewidth=1.2
)

plt.title('Recruitment Funnel Analysis',
          fontsize=18,
          fontweight='bold',
          pad=20)

plt.xlabel('Number of Candidates',
           fontsize=14,
           fontweight='bold')

plt.ylabel('Recruitment Stage',
           fontsize=14,
           fontweight='bold')

ax.xaxis.set_major_formatter(FuncFormatter(lambda x, _: f'{int(x):,}'))

for container in ax.containers:
    labels = [f'{int(v):,}' for v in container.datavalues]
```

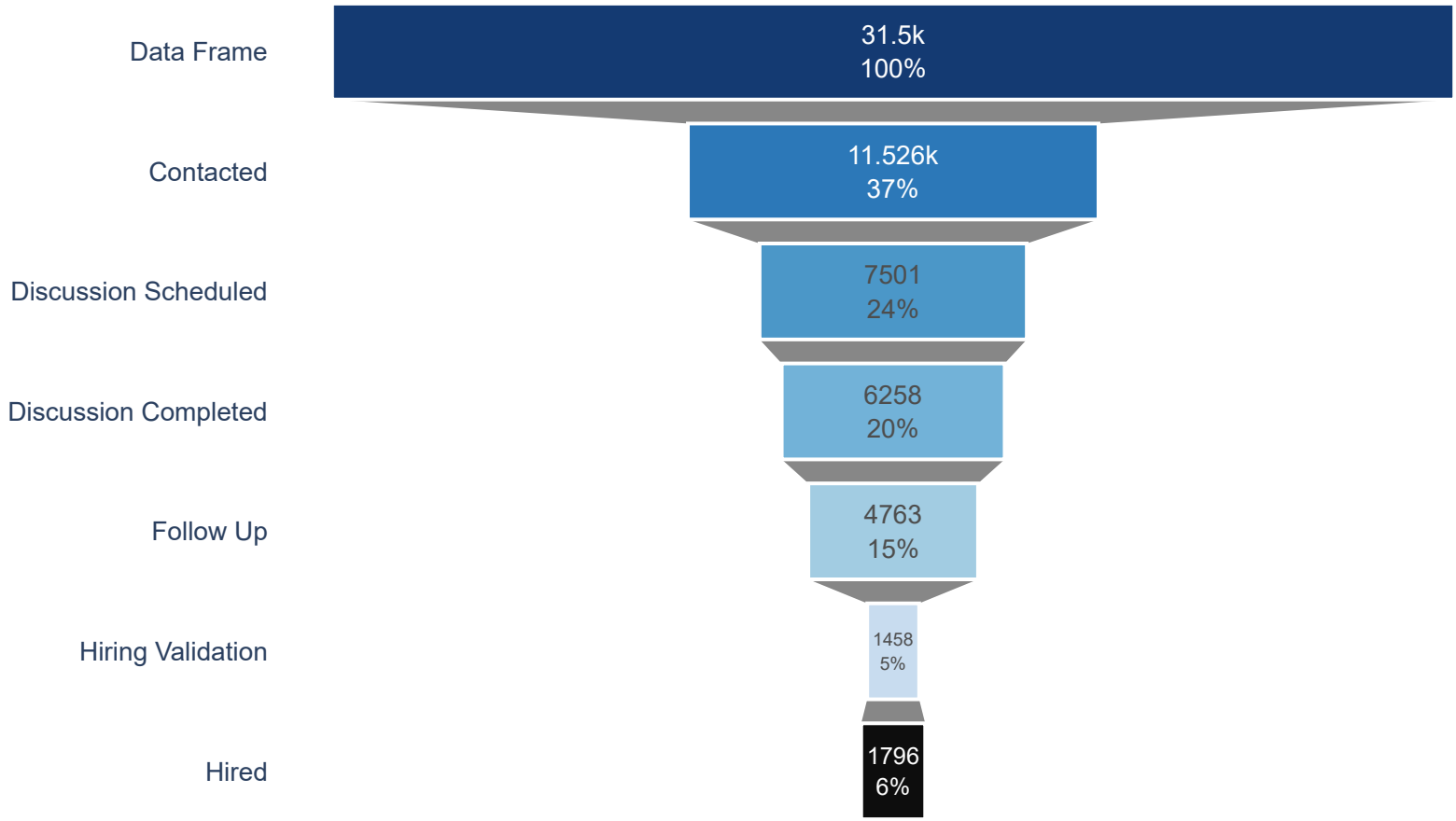
```
    ax.bar_label(
        container,
        labels=labels,
        padding=5,
        fontsize=10,
        fontweight='bold'
    )

ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.grid(axis='x', linestyle='--', alpha=0.35)

plt.tight_layout()
plt.show()
```



In [52]:
```
# Funnel Chart
import plotly.graph_objects as go

fig = go.Figure(go.Funnel(
    y=funnel_df['Stage'],
    x=funnel_df['Candidates'],
```

```
    textinfo='value+percent initial',
    textposition='inside',
    marker=dict(
        color=[
            '#08306B',
            '#2171B5',
            '#4292C6',
            '#6BAED6',
            '#9ECAE1',
            '#C6DBEF'
        ],
        line=dict(color='white', width=2)
    ),
    opacity=0.95
))

fig.update_layout(
    title={
        'text':'<b>Recruitment Funnel Analysis</b>',
        'x':0.5,
        'font':dict(size=24)
    },
    font=dict(
        family='Arial',
        size=14
    ),
    plot_bgcolor='white',
    paper_bgcolor='white',
    margin=dict(l=80, r=80, t=90, b=60),
    width=900,
    height=600
)

fig.show()
```

Recruitment Funnel Analysis



The recruitment funnel analysis demonstrated a clear pattern of candidate attrition across the recruitment process. From the initial 31,500 candidates, 11,526 were contacted, 7,501 reached discussion scheduling, 6,258 completed discussions, and 4,763 progressed to follow-up. The most significant bottleneck occurred between Follow Up and Hiring Validation, where only 1,458 candidates

progressed, representing a 30.61% conversion rate and 69.39% drop-off. However, the final stage shows an increase from 1,458 candidates at Hiring Validation to 1,796 candidates recorded as Hired. This increase should be interpreted in the context of the data history, as Hiring Validation was introduced after 2023 and was not consistently maintained in earlier records. Consequently, some candidates who were recorded as Hired may not have a corresponding Hiring Validation record, resulting in the apparent increase at the final stage. Therefore, the funnel highlights both a significant recruitment bottleneck and an important historical data-recording limitation that is considered.

Exploratory Data Analysis (EDA)

In [53]:
```
# Experience vs Hiring Validation

sns.set_theme(style="whitegrid")

plt.figure(figsize=(8,6))

ax = sns.boxplot(
    data=df,
    x='Hiring Validation',
    y='Years of Experience',
    palette='Set2'
)

plt.title('Years of Experience by Hiring Validation',
          fontsize=16,
          fontweight='bold')

plt.xlabel('Hiring Validation',
           fontsize=13,
           fontweight='bold')

plt.ylabel('Years of Experience',
           fontsize=13,
           fontweight='bold')

plt.tight_layout()

plt.show()
```

In [54]:
```
# Licensing Status vs Hiring Validation

licensing_hiring = pd.crosstab(
    df['Licensing Status'],
    df['Hiring Validation']
)
```

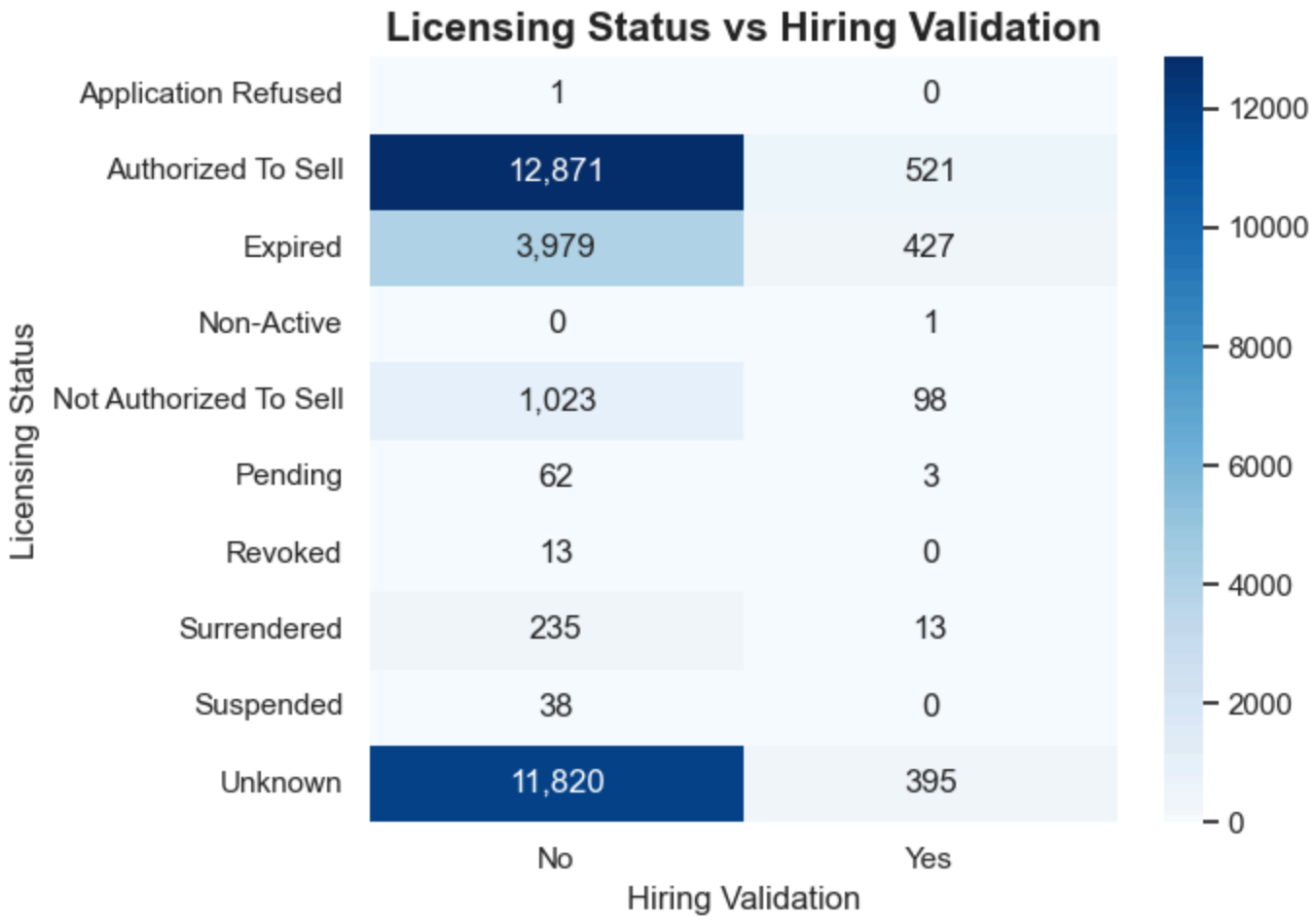
```
licensing_hiring
```

Out[54]:

Hiring Validation	No	Yes
Licensing Status		
Application Refused	1	0
Authorized To Sell	12871	521
Expired	3979	427
Non-Active	0	1
Not Authorized To Sell	1023	98
Pending	62	3
Revoked	13	0
Surrendered	235	13
Suspended	38	0
Unknown	11820	395

In [55]:
```
# Heatmap of Licensing Status vs Hiring Validation
plt.figure(figsize=(7,5))

sns.heatmap(
    licensing_hiring,
    annot=True,
    fmt=',',
    cmap='Blues'
)

plt.title('Licensing Status vs Hiring Validation',
          fontsize=15,
          fontweight='bold')

plt.tight_layout()
```

```
plt.show()
```



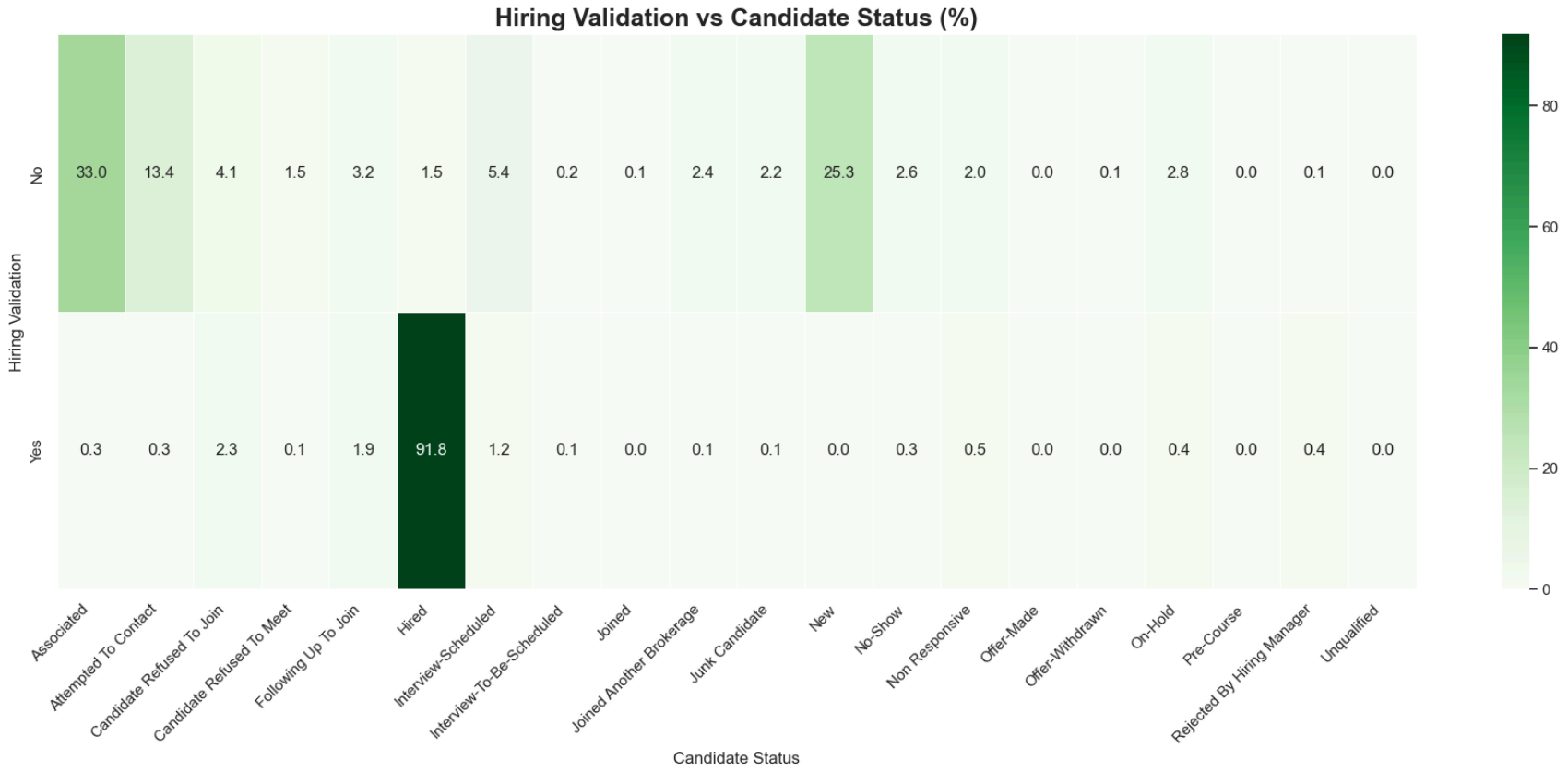
In [56]:
```
# Hiring Validation vs Candidate Status
validation_status = pd.crosstab(
    df['Hiring Validation'],
    df['Candidate Status']
)

validation_status
```

Out[56]:

Candidate Status	Associated	Attempted To Contact	Candidate Refused To Join	Candidate Refused To Meet	Following Up To Join	Hired	Interview-Scheduled	Interview-To-Be-Scheduled	Joined	Joined Another Brokerage	Ju Candida
Hiring Validation											
No	9928	4020	1241	454	955	457	1631	52	38	708	6
Yes	5	5	34	1	28	1339	18	1	0	1	

In [57]:

```
# Heatmap for Hiring Validation vs Candidate Status

discussion_percent = pd.crosstab(
    df['Hiring Validation'],
    df['Candidate Status'],
    normalize='index'
) * 100

plt.figure(figsize=(18,8))

sns.heatmap(
    discussion_percent,
    annot=True,
    fmt='.1f',
    cmap='Greens',
    linewidths=0.5
)

plt.title('Hiring Validation vs Candidate Status (%)',
          fontsize=18,
          fontweight='bold')

plt.xticks(rotation=45, ha='right')

plt.tight_layout()
plt.show()
```



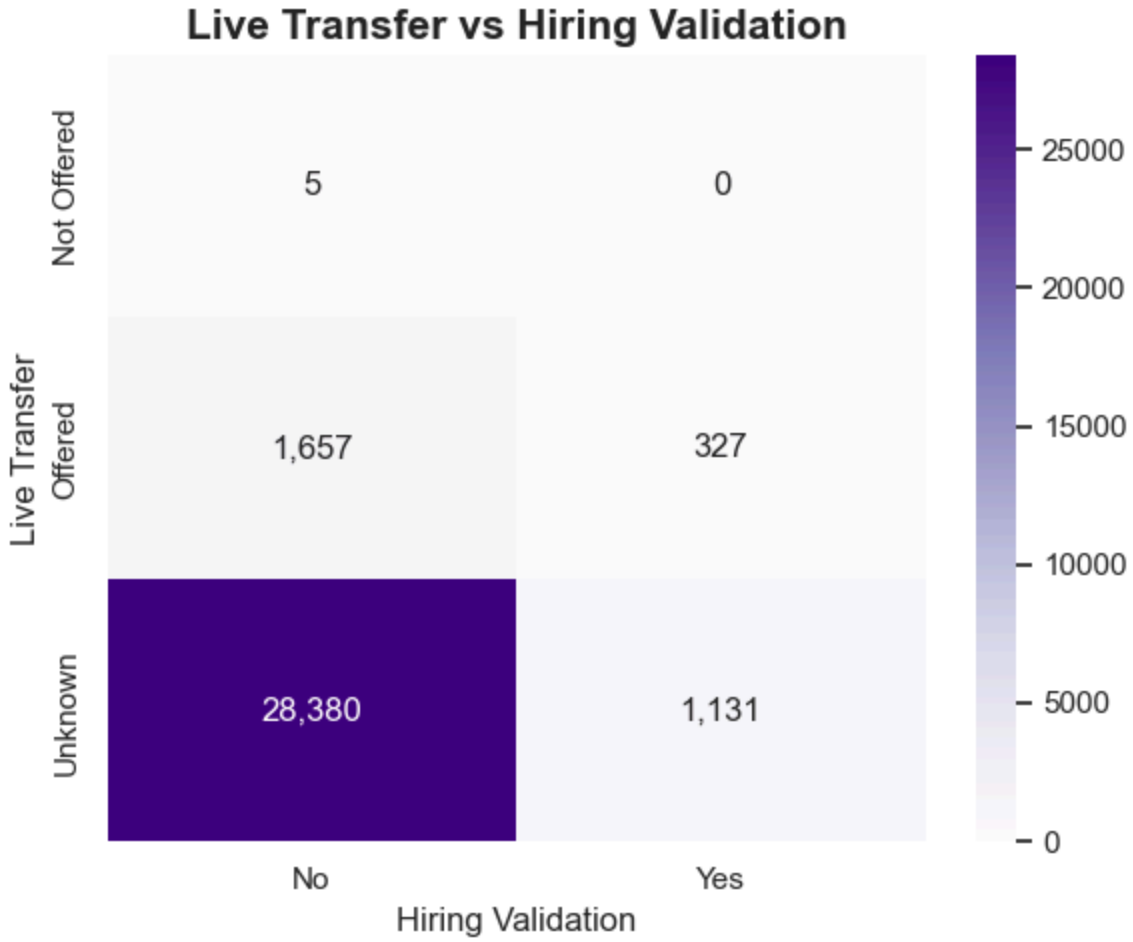
In [58]:
```
# Live Transfer vs Hiring Validation
live_transfer = pd.crosstab(
    df['Live Transfer'],
    df['Hiring Validation']
)

live_transfer
```

Out[58]:

Hiring Validation Live Transfer	No	Yes
Not Offered	5	0
Offered	1657	327
Unknown	28380	1131

In [59]:

```
# Heatmap for Live Transfer vs Hiring Validation
plt.figure(figsize=(6,5))

sns.heatmap(
    live_transfer,
    annot=True,
    fmt=',',
    cmap='Purples'
)

plt.title('Live Transfer vs Hiring Validation',
          fontsize=15,
          fontweight='bold')

plt.tight_layout()

plt.show()
```

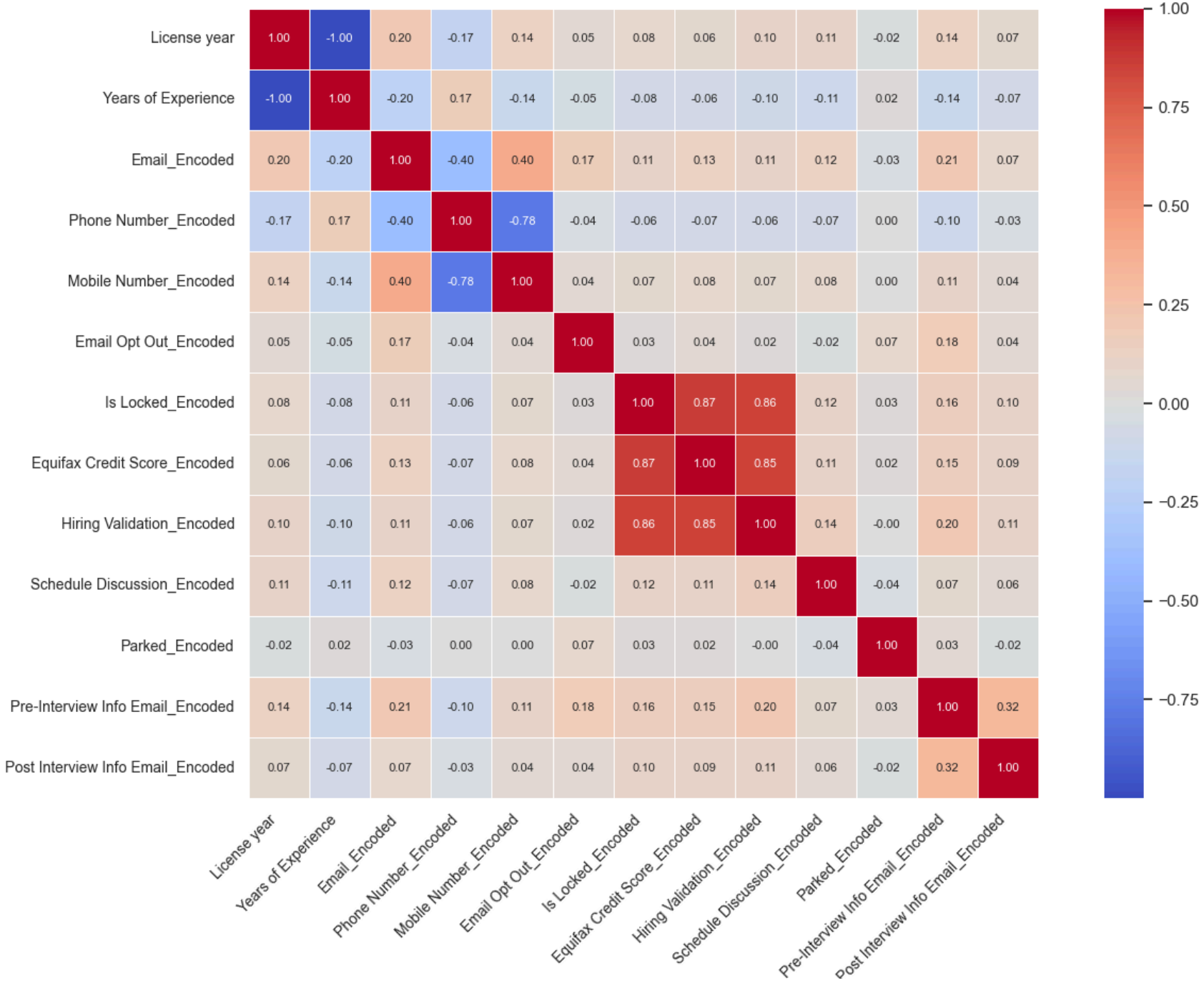


In [60]:
```
# Correlation Analysis

correlation = df.corr(numeric_only=True)
```

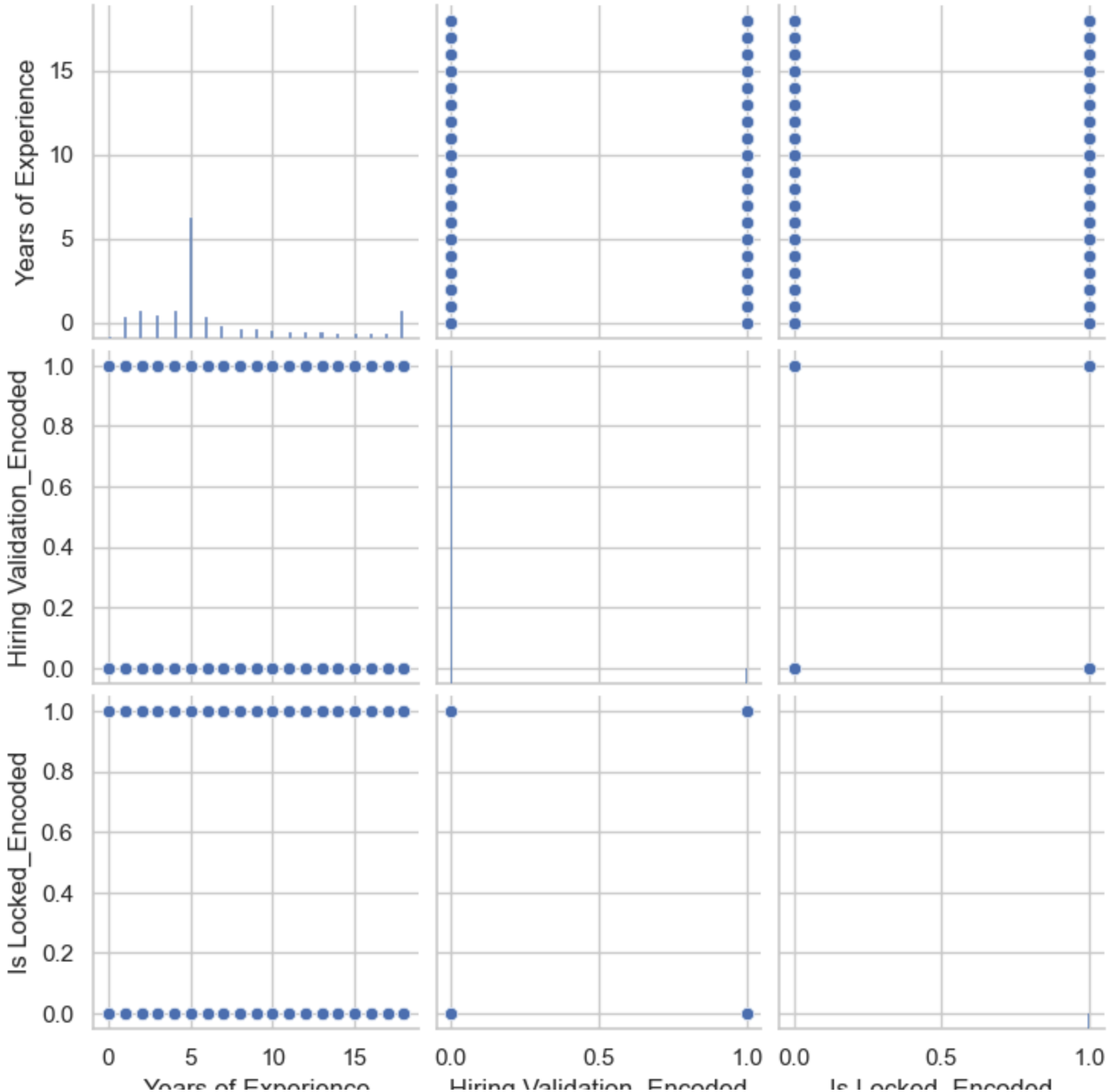
In [61]:
```
# Heatmap of Correlation

plt.figure(figsize=(14,10))

sns.heatmap(
    correlation,
    annot=True,
    fmt=".2f",
```

```
    cmap="coolwarm",
    center=0,
    linewidths=0.5,
    annot_kws={"size":8},
    square=True
)

plt.title(
    "Correlation Matrix",
    fontsize=18,
    fontweight="bold",
    pad=20
)

plt.xticks(rotation=45, ha='right', fontsize=10)
plt.yticks(rotation=0, fontsize=10)

plt.tight_layout()
plt.show()
```

Correlation Matrix



In [62]:
```
sns.pairplot(
    df[
        [
            'Years of Experience',
            'Hiring Validation_Encoded',
            'Is Locked_Encoded'
        ]
    ]
)

plt.show()
```


Years of Experience
Hiring Validation_Encoded
Is Locked_Encoded
0
5
10
15
0.0
0.2
0.4
0.6
0.8
1.0
0.5

Years of Experience Hiring Validation_Encoded Is Locked_Encoded

Overall, the EDA identified several important relationships and patterns within the recruitment data, particularly around Hiring Validation, licensing status, candidate status, years of experience, and live-transfer activity. The analysis indicated that candidates offered a live transfer had a substantially higher validation rate (16.48%) compared with those who were not offered a live transfer (3.83%), while licensing status also showed variation in validation outcomes. These findings provide useful evidence of factors associated with candidate progression; however, they should be interpreted as observed associations rather than causal relationships, as the analysis does not establish that these factors directly cause successful recruitment outcomes.

Feature Engineering

In [63]:
```
# Duplicating the Dataset
df_ml = df.copy()
```

In [64]:
```
df_ml.head()
```

Out[64]:

	Candidate ID.1	Gender	Email	Phone Number	Mobile Number	City	Country	Created Time	Modified Time	Currency	...	Phone Number_Encoded
0	Zr_28480_Cand	Male	Yes	No	Yes	North York	Canada	2023-12-29 09:31:00	2026-03-06 13:01:00	Cad	...	0.0
1	Zr_28476_Cand	Female	Yes	Yes	No	Cambridge	Canada	2023-12-20 16:31:00	2026-05-06 14:43:00	Cad	...	1.0
2	Zr_28475_Cand	Male	Yes	No	Yes	Brampton	Canada	2023-12-20 16:29:00	2026-05-06 14:43:00	Cad	...	0.0
3	Zr_28474_Cand	Male	Yes	No	Yes	Toronto	Canada	2023-12-19 08:54:00	2026-05-06 14:43:00	Cad	...	0.0
4	Zr_28473_Cand	Male	Yes	No	Yes	Cambridge	Canada	2023-12-19 08:54:00	2026-05-06 14:43:00	Cad	...	0.0

5 rows × 56 columns

In [65]:
```
df_ml['Live Transfer'] = df_ml['Live Transfer'].map({
    'Offered':1,
    'Not Offered':0,
    'Unknown':0
})
```

In [66]:
```
df_ml['Gender'] = df_ml['Gender'].map({
    'Male':1,
    'Female':0
})
```

In [67]:
```
encoder = LabelEncoder()
```

In [68]:
```
categorical_columns = [
    'Province',
    'Candidate Status',
    'Candidate Stage',
    'Discussion Stage',
    'Licensing Status',
    'Source',
    'Ethnicity',
    'Religion',
    'Position Category',
    'License Class',
    'Current and Previous Mortgage',
    'Hiring Validation'
]
```

In [69]:
```
for col in categorical_columns:
    df_ml[col] = encoder.fit_transform(df_ml[col].astype(str))
```

In [70]:
```
df_ml = df_ml.drop(columns=[
    'Candidate ID.1',
    'Email',
    'Phone Number',
    'Mobile Number',
    'City',
    'Country',
    'Currency',
    'Last Activity Time',
    'Last emailed',
    'Email Opt Out',
    'Is Hot Candidate',
    'Is Locked',
    'Institute 1 Sign Up Date',
    'Equifax Credit Score',
    'Criminal History',
    'Prior Discipline Action by a Regulator',
    'Professional License Suspension',
    '1 Month Fee Waiver',
    'Work Type',
    'Realtor License',
    'Discussion Type',
    'Schedule Discussion',
```

```
    'Parked',
    'Pre-Interview Info Email',
    'Post Interview Info Email',
    'Created Time',
    'Modified Time',
    'Initial Contact Date',
    'Hire Date',
    'Is Locked_Encoded',
    'Hiring Validation_Encoded'
])
```

In [71]:
```
df_ml.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 31500 entries, 0 to 31499
Data columns (total 25 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Gender                                31500 non-null  int64
 1   Source                                31500 non-null  int64
 2   Candidate Status                      31500 non-null  int64
 3   Ethnicity                             31500 non-null  int64
 4   Religion                              31500 non-null  int64
 5   Position Category                     31500 non-null  int64
 6   License Class                         31500 non-null  int64
 7   License year                          31500 non-null  float64
 8   Years of Experience                   31500 non-null  float64
 9   Current and Previous Mortgage         31500 non-null  int64
 10  Candidate Stage                       31500 non-null  int64
 11  Hiring Validation                     31500 non-null  int64
 12  Licensing Status                      31500 non-null  int64
 13  Province                              31500 non-null  int64
 14  Live Transfer                         31500 non-null  int64
 15  Discussion Stage                      31500 non-null  int64
 16  Email_Encoded                         31500 non-null  float64
 17  Phone Number_Encoded                  31500 non-null  float64
 18  Mobile Number_Encoded                 31500 non-null  float64
 19  Email Opt Out_Encoded                 31500 non-null  float64
 20  Equifax Credit Score_Encoded          31500 non-null  float64
 21  Schedule Discussion_Encoded           31500 non-null  float64
 22  Parked_Encoded                        31500 non-null  float64
 23  Pre-Interview Info Email_Encoded      31500 non-null  float64
 24  Post Interview Info Email_Encoded     31500 non-null  float64
dtypes: float64(11), int64(14)
memory usage: 6.0 MB
```

In [72]:

```
Y = df_ml['Hiring Validation']
X = df_ml.drop('Hiring Validation', axis=1)
print(X.head())
print(Y.head())
```

```
   Gender  Source  Candidate Status  Ethnicity  Religion  Position Category  \
0       1      16                 5         11         4                  2
1       0       7                 9          3         2                  2
2       1       7                 9         11         8                  2
3       1       7                12         11         2                  2
4       1       7                12         11         5                  2

   License Class  License year  Years of Experience  \
0              5        2024.0                  2.0
1              1        2024.0                  2.0
2              5        2021.0                  5.0
3              5        2021.0                  5.0
4              1        2024.0                  2.0

   Current and Previous Mortgage  ...  Discussion Stage  Email_Encoded  \
0                           1886  ...                 2            1.0
1                           1788  ...                 2            1.0
2                           1788  ...                 2            1.0
3                           1788  ...                 2            1.0
4                           1788  ...                 2            1.0

   Phone Number_Encoded  Mobile Number_Encoded  Email Opt Out_Encoded  \
0                   0.0                    1.0                    0.0
1                   1.0                    0.0                    1.0
2                   0.0                    1.0                    1.0
3                   0.0                    1.0                    0.0
4                   0.0                    1.0                    1.0

   Equifax Credit Score_Encoded  Schedule Discussion_Encoded  Parked_Encoded  \
0                           1.0                          0.0             0.0
1                           0.0                          0.0             0.0
2                           0.0                          0.0             0.0
3                           0.0                          0.0             0.0
4                           0.0                          0.0             0.0

   Pre-Interview Info Email_Encoded  Post Interview Info Email_Encoded
0                               0.0                                0.0
1                               1.0                                1.0
2                               1.0                                0.0
3                               1.0                                1.0
4                               1.0                                0.0
```

```
[5 rows x 24 columns]
0    1
1    0
2    0
3    0
4    0
Name: Hiring Validation, dtype: int64
```

In [73]:
```python
y = df_ml['Hiring Validation']
X_original = df_ml.drop('Hiring Validation', axis=1)
print(f"\nOriginal feature set: {X_original.shape[1]} features")
print(f"Target distribution -> Yes: {y.sum():,} ({y.mean()*100:.2f}%), "
      f"No: {(len(y)-y.sum()):,} ({(1-y.mean())*100:.2f}%)")
```

```
Original feature set: 24 features
Target distribution -> Yes: 1,458 (4.63%), No: 30,042 (95.37%)
```

Following feature engineering, the dataset was transformed into a modelling-ready structure by selecting relevant candidate-level variables and converting categorical and binary information into numerical representations. A total of 24 features were retained as predictors, with Hiring Validation defined as the target variable, consisting of 1,458 positive cases (4.63%) and 30,042 negative cases (95.37%). The engineered dataset therefore provided a consistent numerical feature set suitable for the subsequent Logistic Regression, Decision Tree, and Random Forest models, while maintaining the underlying recruitment characteristics captured in the original data

Predictive Analysis

In [74]:
```python
# Spliting the Dataset

X_train, X_test, y_train, y_test = train_test_split(
    X_original, y, test_size=0.20, random_state=RANDOM_STATE, stratify=y
)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

cw = 'balanced'
```

In [75]:
```
print(X_train.shape)
print(X_test.shape)
```

```
(25200, 24)
(6300, 24)
```

In [76]:
```
# Logistic Regression

model = LogisticRegression(max_iter=1000, class_weight=cw)
model.fit(X_train_scaled, y_train)
model_pred = model.predict(X_test_scaled)
model_proba = model.predict_proba(X_test_scaled)[:, 1]

print(classification_report(y_test, model_pred))
print("Confusion matrix:")
print(confusion_matrix(y_test, model_pred))
print(f"PR-AUC: {average_precision_score(y_test, model_proba):.4f}")

lr_original_f1 = f1_score(y_test, model_pred, pos_label=1)
lr_original_acc = accuracy_score(y_test, model_pred)
```

```
              precision    recall  f1-score   support

           0       1.00      0.99      0.99      6008
           1       0.77      0.99      0.86       292

    accuracy                           0.99      6300
   macro avg       0.88      0.99      0.93      6300
weighted avg       0.99      0.99      0.99      6300

Confusion matrix:
[[5920   88]
 [   4  288]]
PR-AUC: 0.9643
```

In [77]:
```
#  Decision Tree

tree = DecisionTreeClassifier(random_state=RANDOM_STATE, class_weight=cw, ccp_alpha=0.0005)
tree.fit(X_train, y_train)
tree_pred = tree.predict(X_test)

print(classification_report(y_test, tree_pred))
```

```
print("Confusion matrix:")
print(confusion_matrix(y_test, tree_pred))

dt_original_f1 = f1_score(y_test, tree_pred, pos_label=1)
dt_original_acc = accuracy_score(y_test, tree_pred)
```

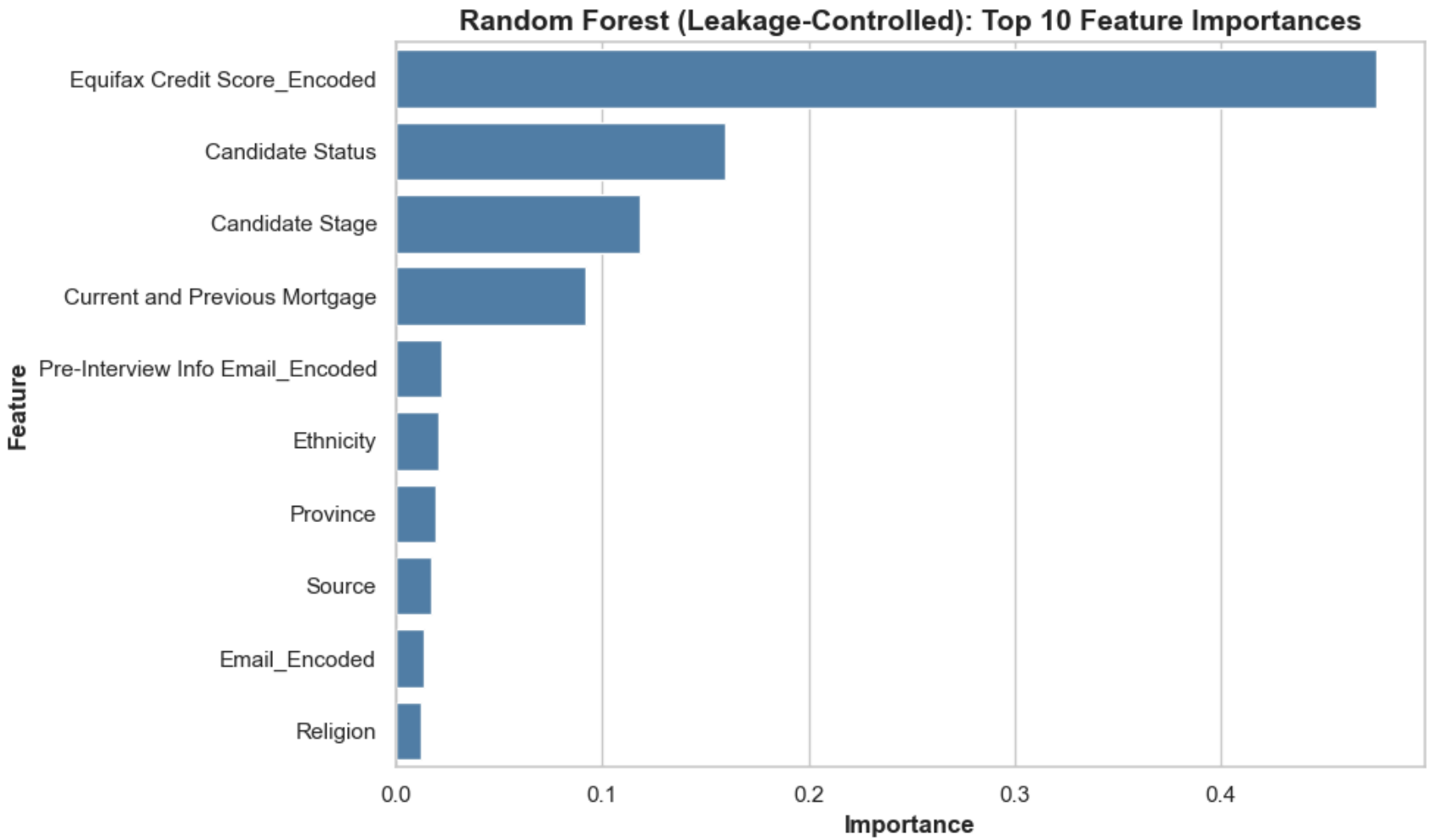
```
              precision    recall  f1-score   support

           0       1.00      0.99      1.00      6008
           1       0.90      0.99      0.94       292

    accuracy                           0.99      6300
   macro avg       0.95      0.99      0.97      6300
weighted avg       0.99      0.99      0.99      6300

Confusion matrix:
[[5975   33]
 [   3  289]]
```

In [78]:
```
#  Random Forest

rf = RandomForestClassifier(random_state=RANDOM_STATE, class_weight=cw, n_estimators=300)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)
rf_proba = rf.predict_proba(X_test)[:, 1]

print(classification_report(y_test, rf_pred))
print("Confusion matrix:")
print(confusion_matrix(y_test, rf_pred))
print(f"PR-AUC: {average_precision_score(y_test, rf_proba):.4f}")

rf_original_f1 = f1_score(y_test, rf_pred, pos_label=1)
rf_original_acc = accuracy_score(y_test, rf_pred)

importance_original = pd.DataFrame({
    'Feature': X_original.columns, 'Importance': rf.feature_importances_
}).sort_values('Importance', ascending=False)
print(importance_original.head(10))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      6008
           1       0.97      0.97      0.97       292

    accuracy                           1.00      6300
   macro avg       0.98      0.98      0.98      6300
weighted avg       1.00      1.00      1.00      6300

Confusion matrix:
[[5998   10]
 [   9  283]]
PR-AUC: 0.9972
                             Feature  Importance
19      Equifax Credit Score_Encoded    0.475356
2                    Candidate Status    0.159813
10                   Candidate Stage    0.117935
9      Current and Previous Mortgage    0.092082
22  Pre-Interview Info Email_Encoded    0.022249
3                           Ethnicity    0.020762
12                          Province    0.019173
1                             Source    0.017029
15                     Email_Encoded    0.013187
4                           Religion    0.011946
```

In [79]:

```python
# Plotting Top 10 Importance

plt.figure(figsize=(10,6))
sns.barplot(
    data=importance_original.head(10),
    x='Importance',
    y='Feature',
    color='steelblue'
)
plt.title('Random Forest (Leakage-Controlled): Top 10 Feature Importances',
          fontsize=14, fontweight='bold')
plt.xlabel('Importance', fontsize=12, fontweight='bold')
plt.ylabel('Feature', fontsize=12, fontweight='bold')
plt.tight_layout()
plt.savefig('feature_importance_leakage_controlled.png', dpi=150)
plt.show()
```



Leakage Controlled

In [80]:
```
leak_risk_cols = ['Candidate Status', 'Candidate Stage', 'Equifax Credit Score_Encoded']
fairness_risk_cols = ['Ethnicity', 'Religion']
redundant_cols = ['License year']  # perfectly correlated with Years of Experience

cols_to_drop = [c for c in (leak_risk_cols + fairness_risk_cols + redundant_cols) if c in X_original.columns]
X_clean = X_original.drop(columns=cols_to_drop)

RANDOM_STATE = 42
```

```
print(f"Leakage-controlled feature set: {X_clean.shape[1]} features "
      f"(removed: {', '.join(cols_to_drop)})")
```

```
Leakage-controlled feature set: 18 features (removed: Candidate Status, Candidate Stage, Equifax Credit Score_Encode
d, Ethnicity, Religion, License year)
```

Predictive Analysis

In [81]:
```
# Spliting the Dataset

X_train2, X_test2, y_train2, y_test2 = train_test_split(
    X_clean, y, test_size=0.20, random_state=RANDOM_STATE, stratify=y
)
scaler2 = StandardScaler()
X_train2_scaled = scaler2.fit_transform(X_train2)
X_test2_scaled = scaler2.transform(X_test2)

cw = 'balanced'
```

In [82]:
```
 # Logistic Regression

lr = LogisticRegression(max_iter=1000, class_weight=cw)
lr.fit(X_train2_scaled, y_train2)
lr_pred = lr.predict(X_test2_scaled)
lr_proba = lr.predict_proba(X_test2_scaled)[:, 1]

print(classification_report(y_test2, lr_pred))
print("Confusion matrix:")
print(confusion_matrix(y_test2, lr_pred))
print(f"PR-AUC: {average_precision_score(y_test2, lr_proba):.4f}")

lr_clean_f1 = f1_score(y_test2, lr_pred, pos_label=1)
lr_clean_acc = accuracy_score(y_test2, lr_pred)
```

```
              precision    recall  f1-score   support

           0       0.99      0.78      0.87      6008
           1       0.15      0.79      0.25       292

    accuracy                           0.78      6300
   macro avg       0.57      0.79      0.56      6300
weighted avg       0.95      0.78      0.84      6300

Confusion matrix:
[[4661 1347]
 [  60  232]]
PR-AUC: 0.2612
```

In [83]:
```python
# Decision Tree

tree2 = DecisionTreeClassifier(random_state=RANDOM_STATE, class_weight=cw, ccp_alpha=0.0005)
tree2.fit(X_train2, y_train2)
tree2_pred = tree2.predict(X_test2)

print(classification_report(y_test2, tree2_pred))
print("Confusion matrix:")
print(confusion_matrix(y_test2, tree2_pred))

dt_clean_f1 = f1_score(y_test2, tree2_pred, pos_label=1)
dt_clean_acc = accuracy_score(y_test2, tree2_pred)
```

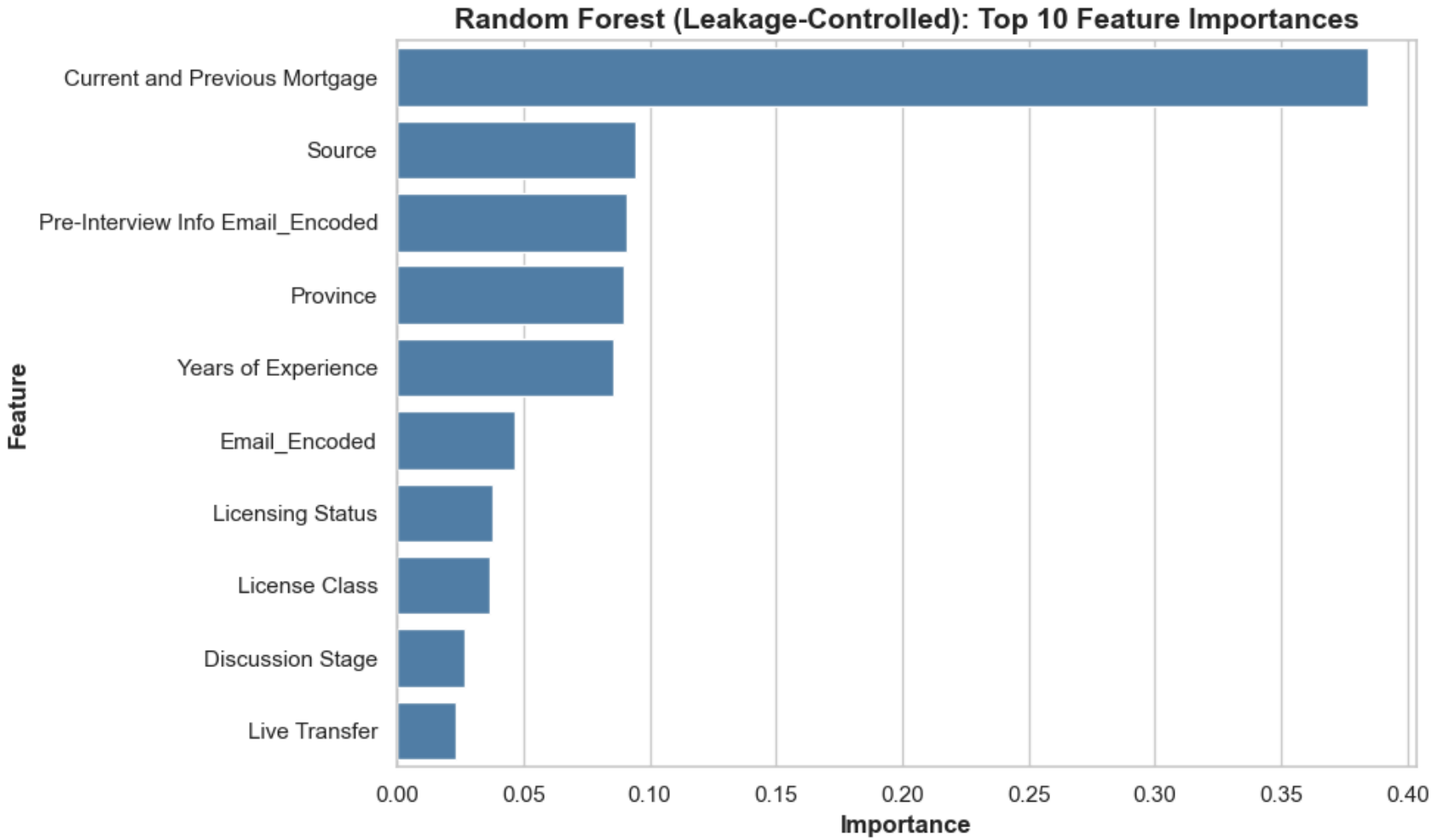
```
              precision    recall  f1-score   support

           0       0.99      0.88      0.93      6008
           1       0.26      0.87      0.40       292

    accuracy                           0.88      6300
   macro avg       0.63      0.87      0.67      6300
weighted avg       0.96      0.88      0.91      6300

Confusion matrix:
[[5283  725]
 [  39  253]]
```

In [84]:
```python
# Random Forest
```

```python
rf2 = RandomForestClassifier(random_state=RANDOM_STATE, class_weight=cw, n_estimators=300)
rf2.fit(X_train2, y_train2)
rf2_pred = rf2.predict(X_test2)
rf2_proba = rf2.predict_proba(X_test2)[:, 1]

print(classification_report(y_test2, rf2_pred))
print("Confusion matrix:")
print(confusion_matrix(y_test2, rf2_pred))
print(f"PR-AUC: {average_precision_score(y_test2, rf2_proba):.4f}")

rf_clean_f1 = f1_score(y_test2, rf2_pred, pos_label=1)
rf_clean_acc = accuracy_score(y_test2, rf2_pred)

importance_clean = pd.DataFrame({
    'Feature': X_clean.columns, 'Importance': rf2.feature_importances_
}).sort_values('Importance', ascending=False)
print(importance_clean.head(10))
```

```
              precision    recall  f1-score   support

           0       0.98      0.98      0.98      6008
           1       0.56      0.61      0.58       292

    accuracy                           0.96      6300
   macro avg       0.77      0.79      0.78      6300
weighted avg       0.96      0.96      0.96      6300

Confusion matrix:
[[5866  142]
 [ 113  179]]
PR-AUC: 0.6610
                            Feature  Importance
5      Current and Previous Mortgage    0.384340
1                            Source    0.094554
16  Pre-Interview Info Email_Encoded    0.090833
7                           Province    0.089680
4                Years of Experience    0.085725
10                     Email_Encoded    0.046678
6                   Licensing Status    0.037618
3                      License Class    0.036353
9                   Discussion Stage    0.026873
8                      Live Transfer    0.023220
```

In [85]:
```
# Plotting Top 10 Importance

plt.figure(figsize=(10,6))
sns.barplot(
    data=importance_clean.head(10),
    x='Importance',
    y='Feature',
    color='steelblue'
)
plt.title('Random Forest (Leakage-Controlled): Top 10 Feature Importances',
          fontsize=14, fontweight='bold')
plt.xlabel('Importance', fontsize=12, fontweight='bold')
plt.ylabel('Feature', fontsize=12, fontweight='bold')
plt.tight_layout()
plt.savefig('feature_importance_leakage_controlled.png', dpi=150)
plt.show()
```



In [86]:
```
# 5-fold Cross-Validation

skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)

rf_cv_original = RandomForestClassifier(random_state=RANDOM_STATE, class_weight='balanced', n_estimators=300)
cv_scores_original = cross_val_score(rf_cv_original, X_original, y, cv=skf, scoring='f1')
print(f"Original spec  - RF 5-fold CV F1: {cv_scores_original.mean():.3f} +/- {cv_scores_original.std():.3f}")

rf_cv_clean = RandomForestClassifier(random_state=RANDOM_STATE, class_weight='balanced', n_estimators=300)
cv_scores_clean = cross_val_score(rf_cv_clean, X_clean, y, cv=skf, scoring='f1')
print(f"Leakage-controlled - RF 5-fold CV F1: {cv_scores_clean.mean():.3f} +/- {cv_scores_clean.std():.3f}")
```

```
Original spec  - RF 5-fold CV F1: 0.969 +/- 0.007
Leakage-controlled - RF 5-fold CV F1: 0.584 +/- 0.043
```

In [87]:
```python
# Comparison table

comparison_df = pd.DataFrame([
    {'Model': 'Logistic Regression', 'Original Accuracy': round(lr_original_acc, 4),
     'Original F1 (minority)': round(lr_original_f1, 4),
     'Leakage-Controlled Accuracy': round(lr_clean_acc, 4),
     'Leakage-Controlled F1 (minority)': round(lr_clean_f1, 4),
     'F1 Drop': round(lr_original_f1 - lr_clean_f1, 4)},
    {'Model': 'Decision Tree', 'Original Accuracy': round(dt_original_acc, 4),
     'Original F1 (minority)': round(dt_original_f1, 4),
     'Leakage-Controlled Accuracy': round(dt_clean_acc, 4),
     'Leakage-Controlled F1 (minority)': round(dt_clean_f1, 4),
     'F1 Drop': round(dt_original_f1 - dt_clean_f1, 4)},
    {'Model': 'Random Forest', 'Original Accuracy': round(rf_original_acc, 4),
     'Original F1 (minority)': round(rf_original_f1, 4),
     'Leakage-Controlled Accuracy': round(rf_clean_acc, 4),
     'Leakage-Controlled F1 (minority)': round(rf_clean_f1, 4),
     'F1 Drop': round(rf_original_f1 - rf_clean_f1, 4)},
])
print(comparison_df.to_string(index=False))
comparison_df.to_csv('leakage_comparison_results.csv', index=False)
importance_clean.to_csv('leakage_controlled_feature_importance.csv', index=False)
```

```
              Model  Original Accuracy  Original F1 (minority)  Leakage-Controlled Accuracy  Leakage-Controlled F1 (m
inority)  F1 Drop
Logistic Regression             0.9854                  0.8623                       0.7767
0.2480   0.6143
      Decision Tree             0.9943                  0.9414                       0.8787
0.3984   0.5429
      Random Forest             0.9970                  0.9675                       0.9595
0.5840   0.3835
```

The predictive analysis initially evaluated Logistic Regression, Decision Tree, and Random Forest using the original 24-feature dataset, with an 80:20 stratified train-test split. The original models produced very strong performance, with minority-class F1-scores of 0.86, 0.94 and 0.97, respectively, and Random Forest achieving the highest overall performance (99.70% accuracy and 0.9675 F1-score). However, feature-importance analysis showed that Candidate Status, Candidate Stage and Equifax Credit Score

were highly influential, raising concerns that the models were using information closely related to the outcome rather than genuinely predictive candidate characteristics.

To address this issue, a leakage-controlled analysis was conducted by removing Candidate Status, Candidate Stage, Equifax Credit Score, as well as Ethnicity, Religion and the redundant License Year variable, reducing the feature set from 24 to 18 predictors. This resulted in a substantial reduction in performance: Logistic Regression achieved 77.67% accuracy and 0.25 F1-score, Decision Tree achieved 87.87% accuracy and 0.40 F1-score, while Random Forest performed best with 95.95% accuracy and 0.58 F1-score. The Random Forest 5-fold cross-validation F1-score also decreased from 0.969 ± 0.007 in the original specification to 0.584 ± 0.043 after leakage control.

The comparison demonstrates that the exceptionally high performance of the original models was partly driven by features that introduced data leakage, rather than reflecting purely prospective predictive capability. Although the leakage-controlled models performed less strongly, their results provide a more realistic assessment of the information available for prediction and therefore offer a more defensible basis for the dissertation's conclusions. Among the three models, Random Forest remained the strongest performer after leakage control, suggesting that the remaining recruitment, communication, licensing and experience variables retain meaningful predictive information.

In [ ]: